# GR-01 : FUNCTIONS

## SINGLE CORRECT

**Q.(1)**

Which of the following function is surjective but not injective

(A) $f : R \to R$ $f(x) = x^4 + 2x^3 - x^2 + 1$

(B) $f : R \to R$ $f(x) = x^3 + x + 1$

(C) $f : R \to R^+$ $f(x) = \sqrt{1+x^2}$

(D) $f : R \to R$ $f(x) = x^3 + 2x^2 - x + 1$

**Q.(2)**

Let $f(x) = \max.\{\sin t : 0 \le t \le x\}$

$g(x) = \min.\{\sin t : 0 \le t \le x\}$

and $h(x) = [f(x) - g(x)]$

where [ ] denotes greatest integer function, then the range of h(x) is-

(A) $\{0,1\}$ (B) $\{1,2\}$ (C) $\{0,1,2\}$ (D) $\{-3,-2,-1,0,1,2,3\}$

## MULTIPLE CORRECT

**Q.(3)**

Let $f : \left(-\frac{\pi}{2}, a\right] \to (-\infty, b]$ be an invertible function with longest domain defined as $f(x) = \tan x + \cot x - 2$ then -

(A) Ordered pair (a,b) is $\left(-\frac{\pi}{3}, -\frac{(4+2\sqrt{3})}{\sqrt{3}}\right)$

(B) Ordered pair (a,b) is $\left(-\frac{\pi}{4}, -4\right)$

(C) $f^{-1}(x) = \tan^{-1}\left(\frac{x+2+\sqrt{x^2+4x}}{2}\right)$

(D) $f^{-1}(x) = \tan^{-1}\left(\frac{x+2-\sqrt{x^2+4x}}{2}\right)$

**Q.(4)**

Which of the following statement(s) is(are) correct ?

(A) If f is a one-one mapping from set A to A, then f is onto.

(B) If f is an onto mapping from set A to A, then f is one-one

(C) Let f and g be two functions defined from $R \to R$ such that gof is injective, then f must be injective.

(D) If set A contains 3 elements while set B contains 2 elements, then total number of functions from A to B is 8.

**Q.(5)**

If $f(x) = x^3 + x^2 + 2x$, $g(x) = e^x$ (where g(x) is invertible function), then which of the following statement(s) is/are correct ?

(A) $gof(x) \le g(x)$ for $x \in (-\infty, 0]$

(B) $g^{-1}of(x) \ge g^{-1}(x)$ for $x \in (0, \infty)$

(C) $g^{-1}of(x) \le g^{-1}(x)$ for $x \in (0, \infty)$

(D) $g^{-1}of(x) \ge g^{-1}(x)$ for $x \in (-\infty, 0]$

**Q.(6)** Let $f: \mathbb{A} \to \mathbb{B}$ and $g: \mathbb{C} \to \mathbb{A}$ be functions. Then which of the following options is/are correct?

(A) If $f$ and $g$ are Injective, then $fog$ is also Injective.

(B) If $f$ and $g$ are Surjective, then $fog$ is also Surjective.

(C) If $fog$ is Injective, then $g$ MUST be Injective.

(D) If $fog$ is Injective, then $f$ need not be Injective.

**Q.(7)** Let $f: \mathbb{A} \to \mathbb{B}$ and $g: \mathbb{C} \to \mathbb{A}$ be functions. Then which of the following options is/are correct?

(A) If $fog$ is Surjective, then $f$ MUST also be Surjective.

(B) If $fog$ is Surjective, then $g$ need not be Surjective.

(C) If $fog$ is Injective and $g$ is Surjective, then $f$ MUST be Injective.

(D) If $f$ and $g$ are Bijective, then $fog$ need not be Bijective.

**Q.(8)** Consider the following sets

$$\mathbb{A} = \left\{ \frac{x^2 - 2x + 4}{x^2 + 2x + 4} \in \mathbb{R} : x \in \mathbb{R} \right\},$$

$$\mathbb{B} = \left\{\frac{x^2 - 2x - 4}{x^2 + 2x - 4} \in \mathbb{R} : x \in \mathbb{R}\right\}.$$

Then which of the following options is/are correct?

(A) $\left\{1, \frac{3}{7}, \frac{7}{3}, 3, \frac{1}{3}, \frac{7}{19}, \frac{19}{7}\right\} \subseteq \mathbb{A}$

(B) $\{-13, 2301, \pi, -e\} \subseteq \mathbb{B}$

(C) $\left\{1, \frac{3}{7}, \frac{7}{3}, 3, \frac{1}{3}, \frac{19}{39}, \frac{39}{19}\right\} \subseteq \mathbb{A}$

(D) $\{-1301, 23, -\pi, e\} \subseteq \mathbb{B}$

**Q.(9)** Which of the following options is/are correct, if

$$f(x) = \max(x^2, |2 - x|) \text{ and } g(x) = \min(x^3, 3x - 2)\,?$$

(A) If $x \in (-\infty, -2]$, then $f(g(x)) = x^6$

(B) If $x \in (-2, 0]$, then $f(g(x)) = 9x^2 - 12x + 4$

(C) If $x \in (0, 1]$, then $f(g(x)) = 4 - 3x$

(D) If $x \in (-2, 1]$, then $g(f(x)) = 4 - 3x$

**Q.(10)** Let $a, b, c \in \mathbb{R}, a + b + c = 3$ and functions $f, g, h$ be such that

$$f : \mathbb{R} \to \left[-\frac{1}{2}, \frac{1}{2}\right]; f(x) = \frac{x}{1 + x^2},$$

$$g(x) = ax^2 + bx + c;\; g(x + y) = g(x) + g(y) + xy \forall x, y \in \mathbb{R} \text{ and}$$

$$h(x) + 2h\left(\frac{1}{x}\right) = 3x \;\forall x \in \mathbb{R} - \{0\}.$$

If $|\mathbb{X}|$ denotes the cardinal number of set $\mathbb{X}$, then which of the following options is/are correct?

(A) $f$ is Injective function.

(B) $f$ is Surjective function.

(C) $g(1) + g(2) + g(3) + \cdots + g(10) = 330$

(D) $|\{x \in \mathbb{R} : h(x) = h(-x)\}| = 2.$

## NUMERICAL TYPE

**Q.(11)**

Let $f : R \to R$ be a function satisfying $f(10 - x) = f(x)$ and $f(2 - x) = f(2 + x) \forall x \in R$. If $f(0) = \lambda$ (where $\lambda \in R$). Then the minimum possible number of values of x satisfying $f(x) = \lambda$ in the interval $[0, 38]$, is

**Q.(12)**

Let A = {1,2,3,4,5} & B = {a,b,c,d}. A function is defined from A to B. 'p' be the number of one-one functions from A to B and 'q' be the number of onto function from A to B. Then (p + q) equals

**Q.(13)** Let

$$f : \mathbb{A} \rightarrow \left[-\frac{1}{2}, 0\right] ; f^3(x) - 3x\, f(x) + x^3 + 1 = 0.$$

If $f$ is an invertible function, then number of solutions of

$$\left(f\left(f(x)\right)\right)^2 = \frac{3}{2} f^{-1}(x) + 1$$

is equal to

**Q.(14)** Let $\lfloor x \rfloor = \max\{a \in \mathbb{Z} : a \leq x\}$ ,$\{x\} = x - \lfloor x \rfloor$,

$$\mathbb{A} = \left\{x \in \mathbb{R} : \frac{\lfloor x+1 \rfloor^2}{6} = \frac{x}{3} + \frac{\{x\}}{2}\right\},$$

$$\mathbb{B} = \left\{x \in \mathbb{R}^+ : \{x^2\} = \left\{\frac{1}{x}\right\}, x^2 \in (2,3)\right\}.$$

If $|\mathbb{X}|$ denotes the cardinal number of set $\mathbb{X}$, then $|\mathbb{A}| + |\mathbb{B}|$ is equal to

**Q.(15)**

If $1 < x^2 < 9$, then number of solutions of the equation $\{|x|\} = \left\{\frac{1}{x^2}\right\}$ (where {.} denote fractional part function) is -

**Q.(16)** Let $|\mathbb{X}|$ denotes the cardinal number of set $\mathbb{X}$ ,

$$\mathbb{A} = \left\{x \in \mathbb{R} : \sqrt{ax^2 + bx} \in \mathbb{R}\right\},$$

$$\mathbb{B} = \left\{\sqrt{ax^2 + bx} \in \mathbb{R} : x \in \mathbb{R}\right\},$$

$$\mathbb{C} = \{a \in \mathbb{R} : \mathbb{A} = \mathbb{B} \text{ for at least one } b \in \mathbb{R}^+\}.$$

Then absolute value of sum of elements of set $\mathbb{C}$ is equal to

**Q.(17)** Number of elements in the set

$$\{5x \in \mathbb{Z} : \log_3(\log_3|\cos x| - \log_{x^2-4x+6}|\cos x|) \in \mathbb{R}\}$$

is equal to

## PASSAGES

**Passage:**

The function $f : R \to R$ is defined as $f(x) = \frac{e^x - e^{-x}}{2}$ and function $g : R \to [1, \infty)$ is defined as $g(x) = \frac{e^x + e^{-x}}{2}$.

On the basis of above information, answer the following questions :

**Q.(18)**

Let $h(x) = \frac{f(x)}{g(x)}$, then $\frac{2h(x)}{1+h^2(x)}$ is equal to

(A) 1 (B) 2 (C) h(2x) (D) $h(x)f(x)$

**Q.(19)**

Number of integers contained in the range of the function $k(x) = \left[\frac{f(x)}{g(x)}\right]$ (where [.] denotes G.I.F.) is

(A) 0 (B) 1 (C) 2 (D) more than 2

## MATRIX MATCHING TYPE

**Q.(20)**

| | Column-I | | Column-II |
|---|---|---|---|
| (A) | Let $f : R \to R$ defined as $f(x) = e^{\text{sgn } x} + e^{x^2}$, where sgn x denotes signum function of x, then f(x) is | (P) | Odd |
| (B) | Let $f : (-1,1) \to R$ defined as $f(x) = x[x^4] + \frac{1}{\sqrt{1-x^2}}$ where [x] denotes greatest integer less than or equal to x, then f(x) is | (Q) | Even |
| (C) | Let $f : R \to R$ defined as $f(x) = \frac{x(x+1)(x^4+1)+2x^4+x^2+2}{x^2+x+1}$ then f(x) is | (R) | Neither odd nor even |
| (D) | Let $f : R \to R$ defined as $f(x) = x + 3x^3 + 5x^5 + \ldots\ldots + 101x^{101}$ then f(x) is | (S) | One-one |
| | | (T) | Many-One |

## KEY

| | | | | |
|---|---|---|---|---|
| 1. D | 2. C | 3. BD | 4. CD | 5. AB |
| 6. ABCD | 7. ABC | 8. ABCD | 9. ACD | 10. BCD |
| 11. 13 | 12. 240 | 13. 1 | 14. 4 | 15. 4 |
| 16. 4 | 17. 9 | 18. C | 19. C | 20. A-RT,B-QT, C-RT,D-PS |

# GR-02 : LIMITS AND CONTINUITY

## SINGLE CORRECT

**Q.(1)**

The value of $\lim\limits_{x\to 0}\dfrac{\ln(\sec(ex)\sec(e^2x).....\sec(e^{50}x))}{e^2-e^{2\cos x}}$ is equal to

(A) $\dfrac{e^{100}-1}{(e^2-1)}$ (B) $\dfrac{e^{100}-1}{2(e^2-1)}$ (C) $\dfrac{2\left(e^{50}-1\right)}{e^2-1}$ (D) $\dfrac{e^2\left(e^{100}-1\right)}{2\left(e^2-1\right)}$

**Q.(2)**

$\lim\limits_{x\to 0}\dfrac{\sin x^4-x^4\cos x^4+x^{20}}{x^4(e^{2x^4}-1-2x^4)}$ is equal to

(A) 0 (B) $-\dfrac{1}{6}$ (C) $\dfrac{1}{6}$ (D) does not exist

**Q.(3)**

Let $g(x) = \tan^{-1}|x| - \cot^{-1}|x|$, $f(x) = \dfrac{[x]}{[x+1]}\{x\}$, $h(x) = |g(f(x))|$ where $\{x\}$ denotes fractional part and $[x]$ denotes the integral part then which of the following holds good ?

(A) h is continuous at $x = 0$ (B) h is discontinuous at $x = 0$
(C) $h(0^-) = \pi/2$ (D) $h(0^+) = -\pi/2$

**Q.(4)** Let $f$ be a positive continuous function satisfying

$$f(x^3-4x)=\frac{2}{f(2^x+7)-1}\forall x\in\mathbb{R}.$$

Also, let $g$ be a continuous function satisfying

$$g(x)=g(2x)\forall x\in\mathbb{R}, g(1)=\frac{\pi}{2},$$

then find

$$\lambda = f(15) + \lim_{x \to 0} \frac{\cos^2 g(x) + 1 - \sin^3 g(x)}{\sin^2 x}.$$

(A) 1 (B) 2 (C) 3 (D) 4

**Q.(5)** Which of the following options is/are correct, if

$$\mathbb{S} : \text{If } a_{n-1} = \frac{a_n - n}{n} \forall n \in \mathbb{N} - \{1\}, a_1 = 1, \text{then } \lim_{n \to \infty} \prod_{r=1}^{n} \left(1 + \frac{1}{a_r}\right) = e.$$

$$\mathbb{T} : \lim_{n \to \infty} \left(\frac{n^2 + n + 1}{n + 1} - (an + b)\right) = 4 \Longrightarrow 7a - 2b = 15.$$

(A) $\mathbb{S}$ is True and $\mathbb{T}$ is False

(B) $\mathbb{S}$ is False and $\mathbb{T}$ is True

(C) Both $\mathbb{S}$ and $\mathbb{T}$ are True

(D) Both $\mathbb{S}$ and $\mathbb{T}$ are False.

## MULTIPLE CORRECT

**Q.(6)**

Which of the following limits are non existent ?

(A) $\lim_{x \to 0} \frac{1}{x} \sqrt{\frac{1-\cos x}{1+\cos x}}$

(B) $\lim_{x \to 0} \frac{|x|}{x}$

(C) $\lim_{x \to 0} \frac{\cot^{-1} \frac{1}{x}}{x}$

(D) $\lim_{x \to \pi/2} \cos^{-1}[\cot x]$ where [.] is greatest integer function

**Q.(7)**

Which of the following is/are correct :

(A) $\lim_{x\to 0}\frac{\sin 4x}{1-\sqrt{1-x}}=8$

(B) $\lim_{x\to \pi/2}\frac{\cos 3x+3\cos x}{\left(\frac{\pi}{2}-x\right)^3}=2$

(C) $\lim_{x\to \pi/6}\frac{2-\sqrt{3}\cos x-\sin x}{(6x-\pi)^2}=\frac{1}{36}$

(D) $\lim_{x\to \pi}\frac{1-\sin\frac{x}{2}}{\cos\frac{x}{2}\left(\cos\frac{x}{4}-\sin\frac{x}{4}\right)}=\frac{1}{2}$

**Q.(8)**

If $L=\lim_{x\to 1}\frac{(1-x)(1-x^2)(1-x^3)........(1-x^{2n})}{[(1-x)(1-x^2)(1-x^3)........(1-x^n)]^2}$ then L can be equal to

(A) $\prod_{r=1}^{n}\frac{n+r}{r}$

(B) $\frac{1}{n!}\prod_{r=1}^{n}(4r-2)$

(C) The sum of the coefficients of two middle terms in the expansion of $(1+x)^{2n-1}$.

(D) The coefficient of $x^n$ in the expansion of $(1+x)^{2n}$.

**Q.(9)**

Let $f(x)=\frac{\cos^{-1}(\cos x)}{x}$, then which of the following is(are) correct (where [.] denotes greatest integer function) ?

(A) $\lim_{x\to 0^+}[f(x)]=1$ (B) $\lim_{x\to 0^-}[f(x)]=1$ (C) $\lim_{x\to 0^+}[f(x)]=0$ (D) $\lim_{x\to 0^-}[f(x)]=-1$

**Q.(10)**

Let $f(x)=\begin{cases}|x|, & -2\le x\le 0\\ e^x-1, & 0<x\le 5\end{cases}$, $g(x)=\begin{cases}-x-1, & -1\le x\le 0\\ \dfrac{4x-3}{3}, & 0<x\le 3\end{cases}$, then -

(A) $\lim\limits_{x\to 0} f(g(x))=1$

(B) $\lim\limits_{x\to 0^-} f(g(2^x))=1$

(C) $\lim\limits_{x\to 0^+} f(g(2^x))=e^{1/3}-1$

(D) $\lim\limits_{x\to 0^+} f(g(2^x-1))=\lim\limits_{x\to 0^-} f(g(2^x-1))$

**Q.(11)** Let $\lfloor x\rfloor=\max\{a\in\mathbb{Z}: a\le x\}$, Then which of the following options is/are correct?

(A) If $f$ is a continuous function such that $\lfloor f(-1)\rfloor=2, \lfloor f(0)\rfloor=-2$, then $f(x)=0$ has at least one solution in $(-1,0)$.

(B) If $f$ is a dis-continuous at $x=a$, and $g$ is continuous at $x=a, g(a)=0$, then $f(x)g(x)$ must be continuous at $x=a$.

(C) If $f(x)+\sin x$ is continuous and $g(x)-e^x$ is dis-continuous at $x=a$, then $f(x)g(x)$ must be dis-continuous at $x=a$.

(D) It is possible that $f(x)$ and $g(x)=\max\{t\in\mathbb{Z}: t\le f(x)\}$ are both continuous at $x=a, f(a)\in\mathbb{Z}$.

**Q.(12)** Which of the following options is/are correct, if

$$f(x)=\lim_{n\to\infty}\frac{\sin\pi x-x^{2n}\sin(x-1)}{1+x^{2n+1}-x^{2n}};n\in\mathbb{N},\text{and}$$

$$\mathbb{A}=\{a\in\mathbb{Z}: g(x)=\lfloor x\rfloor^2-\lfloor x^2\rfloor \text{ is continuous at x = a.}\}$$

Here $|\mathbb{X}|$ denotes the cardinal number of set $\mathbb{X}$

(A) $\lim\limits_{x\to 1^+} f(x)=-1$

(B) $\lim\limits_{x\to 1^-} f(x)=-1$

(C) $f$ is dis-continuous at $x=1$.

(D) $|\mathbb{A}|=0$.

**Q.(13)** Let

$$\lambda = \lim_{x\to 0}\left(\frac{\tan x \cdot \cos\sin x - \sin\sin x}{\tan x \cdot \cos\tan x - \sin\tan x}\right), \mu = \lim_{x\to 0}\left(\frac{a-\sqrt{a^2-x^2}-\frac{x^2}{4}}{x^4}\right); a > 0.$$

If $\mu$ is finite, then

(A) $a-2\lambda = 3$ (B) $64\mu + a = 3$ (C) $a - 2\lambda = 2$ (D) $32\mu = 1$

**Q.(14)** Let $\mu_1 = \lim_{a\to 0^+}(p), \mu_2 = \lim_{a\to 0^+}(q)$, where $p, q$ are roots of

$$\left(\sqrt[3]{1+a}-1\right)x^2 + \left(\sqrt{1+a}-1\right)x + \left(\sqrt[6]{1+a}-1\right) = 0; p > -1, \text{and}$$

$$\lambda_1 = \lim_{x\to\infty} g(x), \lambda_2 = \lim_{x\to-\infty} g(x); \text{where } g(x) = \frac{\ln(x^2+e^x)}{\ln(x^4+e^{2x})},$$

If $\mu_1 > \mu_2$,then which of the following options is/are correct?

(A) $2\lambda_1 - \mu_2 = 2$ (B) $4\lambda_2 - 2\mu_1 = 3$ (C) $3\lambda_1 - \mu_1 = 2$ (D) $3\lambda_2 - 5\mu_2 = 3$

**Q.(15)** Let functions $f$ and $g$ are defined as

$$f(x) = \lim_{h\to 0}\frac{\sin(x+3h) - 3\sin(x+2h) + 3\sin(x+h) - \sin x}{h(\tan(x+2h) - 2\tan(x+h) + \tan x)} \text{ and}$$

$$g(x) = \frac{1-x(1+|1-x|)}{|1-x|} f\left(\frac{1}{1-x}\right)\sin\frac{1}{1-x}\sec^3\frac{1}{1-x}.$$

Then which of the following options is/are correct?

(A) $\lim_{x\to 1^+} g(x)$ does not exists
(B) $\lim_{x\to 1^-} g(x)$ does not exists
(C) $\lim_{x\to 1^+} g(x) = 0$
(D) $\lim_{x\to 1^-} g(x) = 0$

## NUMERICAL TYPE

**Q.(16)**

If the value of limit $\lim_{x\to 0}\dfrac{(1+x)^{\frac{1}{x}}-e+\dfrac{ex}{2}}{x^2}=\dfrac{Ae}{B}$, where A and B are co-prime natural numbers, then find the value of (A+B) :-

**Q.(17)** Find the value of $(a-2be+c)\in\mathbb{R}$, if $b,c\in\mathbb{R}-\{0\}$,

$$\lim_{x\to 0^+}\frac{(1-x)^{\frac{1}{x}}-\frac{1}{e}}{x^a}=b \text{ and } \lim_{x\to\frac{\pi}{2}}\left(\frac{4\sqrt{2}(\sin x+\sin 3x)}{2\sin 2x\sin\frac{3x}{2}+\cos\frac{5x}{2}-\sqrt{2}-\sqrt{2}\cos 2x-\cos\frac{3x}{2}}\right)=c.$$

Here, $e$ denotes the base of natural logarithm.

**Q.(18)** If

$$f(x)=\begin{cases}\dfrac{a^{\sin x}-a^{\tan x}}{\tan x-\sin x}, & x>0\\ k, & x=0\\ \dfrac{\ln(1+x+x^2)+\ln(1-x+x^2)}{\sec x-\cos x}, & x<0\end{cases}$$

is continuous at $x=0$, and

$$g(x)=\ln\left(2-\frac{x}{a}\right)\cot(x-a)\,;x\neq a,a>0,$$

is continuous at $x=a$, then $\left\lfloor g\left(\frac{1}{e}\right)+2e\right\rfloor$ is equal to(where $\lfloor x\rfloor=\max\{a\in\mathbb{Z}:a\leq x\}$)

**Q.(19)** Find the value of $\lambda+n$, where

$$\lim_{x\to 0}\left(\frac{\cos^2 x-\cos x-e^x\cos x+e^x-\frac{x^3}{2}}{x^n}\right)=\lambda(\in\mathbb{R}-\{0\});n\in\mathbb{N}.$$

**Q.(20)** Let

$$\mathbb{A}=\left\{a\in\mathbb{W}:\lim_{x\to 1}\left(\frac{-ax+\sin(x-1)+a}{x+\sin(x-1)-1}\right)^{\frac{1-x}{1-\sqrt{x}}}=\frac{1}{4}\right\},k=\lim_{x\to 0^+}\left(\frac{e^{x^x-1}-x^x}{(x^{2x}-1)^2}\right).$$

Then $\max(\mathbb{A})+8k$ is equal to

## PASSAGES

**Passage-I**

Let

$$f(x)=\begin{cases}\dfrac{e^{\frac{1}{x-1}}-2}{e^{\frac{1}{x-1}}+2}, & x\neq 1\\ 1, & x=1\end{cases} \text{ and } g(x)=\begin{cases}a_n+\sin\pi x\,, & x\in[2n,2n+1]\\ b_n+\cos\pi x, & x\in(2n-1,2n)\end{cases};n\in\mathbb{Z}.$$

Now, based on above information, answer the following questions

**Q.(21)** Which of the following options is/are correct?

(A) $\lim\limits_{x\to1^+} f(x)=1$ (B) $\lim\limits_{x\to1^-} f(x)=-1$ (C) $\lim\limits_{x\to1} f(x)=1$ (D) $\lim\limits_{x\to1} f(x)=-1$

**Q.(22)** If $g$ is continuous for all $x\in\mathbb{R}$, then

(A) $a_{n-1}=b_{n-1}$ (B) $a_n-b_n=1$ (C) $a_n-b_{n+1}=1$ (D) $a_{n-1}-b_n=-1$

**Passage-II**

Let $f$ and $g$ are continuous functions defined from $[0,1]\to\mathbb{R}$ satisfying

$$\max\{f(x):x\in[0,1]\}=\max\{g(x):x\in[0,1]\}.$$

Also $h(x)=x\cos(\pi x+\pi\lfloor x\rfloor)$, where $\lfloor x\rfloor=\max\{a\in\mathbb{Z}:a\leq x\}$.

Now, based on above information, answer the following questions

**Q.(23)** Which of the following options is/are correct?

(A) $(f(c))^2+3f(c)=(g(c))^2+3g(c)$ for some $c\in[0,1]$
(B) $(f(c))^2+f(c)=(g(c))^2+3g(c)$ for some $c\in[0,1]$
(C) $(f(c))^2+3f(c)=(g(c))^2+g(c)$ for some $c\in[0,1]$
(D) $(f(c))^2=(g(c))^2$ for some $c\in[0,1]$

**Q.(24)** The function $h$ is dis-continuous at $x =$

(A) $-1$ (B) 0 (C) 2 (D) 1

## MATRIX MATCHING TYPE

**Q.(25)**

Let $f(x) = \begin{cases} \dfrac{e^{(a+2)x}-1}{x} & , \ x<0 \\ c & , \ x=0 \\ \dfrac{(x+b(x^4))^{1/2}-x^{1/2}}{b(x^{7/2})} & , \ x>0 \end{cases}$ is continuous function $\forall x \in R$

| | **Column-I** | | **Column-II** |
|---|---|---|---|
| (A) | Complete set of value(s) of a is | (P) | $R^+$ |
| (B) | Complete set of value(s) of b is | (Q) | $\left\{-\frac{3}{2}\right\}$ |
| (C) | Complete set of value(s) of c is | (R) | $\left\{\frac{1}{2}\right\}$ |
| (D) | Complete set of value(s) of 2ac is | (S) | $R - \{0\}$ |
| | | (T) | $\left\{-\frac{3}{2}, \frac{1}{2}\right\}$ |

## KEY

| | | | | |
|---|---|---|---|---|
| 1. B | 2. C | 3. A | 4. B | 5. C |
| 6. ABCD | 7. AC | 8. ABCD | 9. AD | 10. ACD |
| 11. AD | 12. AC | 13. AB | 14. ABC | 15. AD |
| 16. 35 | 17. 10 | 18. 2 | 19. 4.5 | 20. 1 |
| 21. AB | 22. BD | 23. AD | 24. ACD | 25. A-Q,B-P, C-R,D-Q |

# GR-03 : DIFFERENTIABILITY AND DIFFERENTIATION

## SINGLE CORRECT TYPE QUESTIONS

**Q.(1)**

If $f(x)=\begin{cases} \ln(1+|x|+(b-2)|x+1|)+a\tan^{-1}x+2 & , \quad -2<x<0 \\ b & , \quad x=0 \\ \dfrac{(c+2)\sin^{-1}\sqrt{\{x\}}+e^{2x}-e^{\ln\left(4\left\{\frac{x}{2}\right\}\right)}-1}{x^2} & , \quad 0<x<2 \end{cases}$ is derivable in (–2, 2) then find the value of $(9a^2+b^2+c^2)$

[Note : $\{y\}$ denotes fractional part of y]

(A) 60 (B) 40 (C) 57 (D) 58

**Q.(2)** Let $f_1$ be a differentiable function satisfying

$$f_2(x)=\lim_{n\to\infty}\binom{n}{k}\left(f_1\left(\frac{x}{n}\right)-f_1(0)\right)^k; \qquad f_1'(0)=\frac{1}{3},$$

$$f_3(k)=\lim_{x\to 0}\frac{1}{x}\sum_{r=1}^{k}f_3\left(\frac{x}{r}\right); \qquad k\in\mathbb{N}, f_3(0)=0, f_3'(0)=1.$$

Then which of the following option(s) is/are equal to

$$\lambda=\sum_{r=0}^{10}\left(\binom{10}{r}\cdot r!\cdot f_2(3)+\frac{f_3(r+1)-f_3(r)}{f_3(11)}\right).$$

(A) 1024 (B) 1026 (C) 1025 (D) 1203

**Q.(3)**

Let $[\![f(x):\mathbb{A}]\!]$ denote the number of points on non-differentiability of any function $y=f(x)$ in the set $\mathbb{A}$. If $g$ is a polynomial function satisfying

$$g(x+y) = g(x) + g(y) + 2xy - 1 \forall x \in \mathbb{R}, \quad f'(0) = 3,$$

then which of the following option(s) is/are equal to

$$\lambda = \left[\!\left[ \lim_{n\to\infty} \sqrt[n]{\sin^{2n} x + \cos^{2n} x} : [-2\pi, 2\pi] \right]\!\right] + \left[\!\left[ \left| |g(x) - 1| - 1 \right| : \mathbb{R} \right]\!\right].$$

(A) 14 (B) 15 (C) 16 (D) 17

## MULTIPLE CORRECT TYPE QUESTIONS

**Q.(4)**

$f(x) = \begin{cases} x+a & \text{if } x<0 \\ |x-1| & \text{if } x \ge 0 \end{cases}$ and $g(x) = \begin{cases} x+1 & \text{if } x<0 \\ (x-1)^2 + b & \text{if } x \ge 0 \end{cases}$. Where a and b are non negative real numbers.

If (gof) (x) is continuous for all real x.

(A) $2a + 3b = 5$ (B) $3a - 2b = 3$

(C) g is differentiable at 0 (D) g is not differentiable at 0.

**Q.(5)**

The function f defined by $f(x) = \begin{cases} x[x], & 0 \le x < 2 \\ (x-1)[x], & 2 \le x \le 3 \end{cases}$ where [x] = greatest integer less than or equal to

1, is not

(A) continuous at 1 (B) continuous at 2

(C) differentiabile at 1 (D) differentiabile at 2.

**Q.(6)** The derivative of

$$f(x) = \tan^{-1}\frac{2x}{1-x^2} \text{ w.r.t. } g(x) = \sin^{-1}\frac{2x}{1+x^2}$$

is equal to

(A) $1\ at\ x = -e$ (B) $-1\ at\ x = -e^{-1}$ (C) $1\ at\ x = e^{-1}$ (D) $-1\ at\ x = e$

**Q.(7)** Which of the following options is/are correct, for

$$f(x) = \begin{cases} x\left(\dfrac{a^{\frac{1}{x}} - a^{\frac{-1}{x}}}{a^{\frac{1}{x}} + a^{\frac{-1}{x}}}\right), & x \neq 0 \\ 0, & x = 0 \end{cases}.$$

(A) If $a \in (0,1)$, then $f$ is continuous but not differentiable

(B) If $a = 1$, then $f$ is continuous but not differentiable

(C) If $a > 1$, then $f$ is continuous but not differentiable

(D) If $a \in \mathbb{R}$, then $f$ is continuous but not differentiable

**Q.(8)** Let $f : [0, \infty) \to [0, \infty)$; $f(x) = xe^{-f(x)}$ be a bijective function. Then which of the following options is/are correct, if

$$\lambda = \max\{a \in \mathbb{Z} : a \le f(f(e^{e+1}))\}, \quad \text{and} \quad \mu = \lim_{n\to\infty} \frac{f(x)}{\ln x}?$$

(A) $\lambda + \mu = 2$ (B) $3\lambda + 2\mu = 4$ (C) $5\lambda - \mu = 4$ (D) $4\lambda + 5\mu = 10$

**Q.(9)** Let $\lfloor x \rfloor = \max\{a \in \mathbb{Z} : a \le x\}$, $\{x\} = x - \lfloor x \rfloor$,

$$f(x) = \lfloor x \rfloor^{\{x\}} + \{x\}^{\lfloor x \rfloor}; f'\left(\frac{5}{2}\right) = 1 + \sqrt{\lambda}\ln 2, g(x) = \begin{cases} b\sin^{-1}\dfrac{x+c}{2}, & -\dfrac{1}{2} < x < 0 \\ \dfrac{1}{2}, & x = 0 \\ \dfrac{e^{\frac{ax}{2}} - 1}{x}, & 0 < x < \dfrac{1}{2} \end{cases}.$$

If $g$ is differentiable at $x = 0$ and $|c| < \frac{1}{2}$, then

(A) $a(4 - c^2) = \lambda^6 b^2$ (B) $a(4 - c^2) = \lambda^{12} b^2$

(C) $a(\lambda^2 - c^2) = \lambda^6 b^2$ (D) $a(4 - c^2) = \lambda^3 b^2$

**Q.(10)** Let $f_n(\alpha)$ denote the $n^{th}$ derivative of function $f$ at $x = \alpha$. Also

$$f(x) = -\frac{x^3}{3} + x^2 \sin\frac{3a}{2} - x\sin a \sin 2a - 5\sin^{-1}(a^2 - 8a + 17),$$

$$g(x) = h\left(x - \sqrt{1 - x^2}\right), h'(x) = 1 - x^2 \text{ and } p(x) = \frac{3x - 1}{x^2 - 1}.$$

Then, which of the following options is/are correct?

(A) $f'(\sin 8) < 0$ (B) $25\, g'\left(\frac{3}{5}\right) = 42$ (C) $p_{50}(0) = 50!$ (D) $f'(\sin 8) > 0$

## NUMERICAL ANSWER TYPE QUESTIONS

**Q.(11)**

If $\sqrt{4\cos x - 3\sin x + 5 + y\sin x} = y\cos 2x$, then $y'(\pi)$ is equal to

**Q.(12)**

If $f(x) = x + 1$, $g^{-1}(x) = x^3 + x + 1$ and $g(f(x)) = h(x)$, then $\lim\limits_{x\to 1}\dfrac{h^{-1}(x) - h^{-1}(1)}{x-1}$ is equal to -

**Q.(13)**

Let $f(x) = 3x^{10} - 7x^8 + 5x^6 - 21x^3 + 3x^2 - 7$ then $265\left[\lim\limits_{h\to 0}\dfrac{h^4 + 3h^2}{(f(1-h) - f(1))\sin(5h)}\right]$ equals

**Q.(14)**

Let f be a function that is differentiable every where and that has the following properties:

(i) $f(x + h) = f(x)\cdot f(h)$ (ii) $f(x) > 0$ for all real x. (iii) $f'(0) = -1$

Then find $\sum\limits_{r=1}^{10}\ell n\left(\dfrac{1}{f(r)}\right)$

**Q.(15)** Find

$$\lambda = \max\{a \in \mathbb{Z} : a \le \max(\mathbb{A})\}, \text{ where}$$

$$\mathbb{A} = \left\{|12a + 4b + c| \;\middle|\; |ax^3 + bx^2 + cx + 5| \le |e^x - e^2| \forall x \ge 0\right\}.$$

## PASSAGES

**Passage-I:**

Let $f$ be a differentiable function satisfying

$$\lim_{y\to 0}\left(\frac{f(x+y)}{f(x)}\right)^{\frac{1}{y}} = e^{f(-x)\tan x}.$$

Also right hand derivative at $x = 0$ exists for $h$,satisfying

$$g(x-y) = g(x)g(y) + h(x)h(y) \forall x, y \in \mathbb{R}.$$
$$and$$
$$h(x-y) = h(x)g(y) - g(x)h(y) \forall x, y \in \mathbb{R}.$$

Now, based on above information, answer the following questions

**Q.(16)** The value of $f'(0)$ is equal to

**Q.(17)** The value of $g'(0)$ is equal to

**Passage-II:**

The function $f$ satisfies the following conditions:

$$f(x) = \begin{cases} ax(x-1)+b, & x < 1 \\ x-1, & 1 \le x \le 3; \\ p(x)^2 + qx + 2, & x > 3 \end{cases} \quad \begin{matrix} f \text{ is continuous for all } x \in \mathbb{R}, \\ f'(1)\ d.n.e., \\ f'(x) \text{ is continuous at } x = 3. \end{matrix}$$

Further given that function $g$ is differentiable at $x = 0$,

$$g(x) = (\cos x)^{\frac{1}{x}} \quad \text{and} \quad \lambda = \lim_{n\to\infty}\left(\frac{2}{\pi}(n+1)\cos^{-1}\frac{1}{n} - \left(n + \frac{2}{\pi}\right)\right).$$

Now, based on above information, answer the following questions.

**Q.(18)** The value of $b + p + |q|$ is equal to

**Q.(19)** The value of $\lambda - 2g'(0)$ is equal to

## MATRIX MATCHING

**Q.(20)** Consider the following

$$f(x) = g(g(x)); \quad \text{where} \quad g(x) = \begin{cases} 1-x, & x \in [0,1] \\ x+2, & x \in (1,2). \\ 4-x, & x \in [2,4] \end{cases}$$

Now, Match the following columns for function $f$

Column-I　　　　Column-II

| | |
|---|---|
| (A) $x = 0$ | (P) continuous |
| (B) $x = 1$ | (Q) dis-continuous |
| (C) $x = 2$ | (R) differentiable |
| (D) $x = 3$ | (S) non-differentiable |

## KEY

| | | | | |
|---|---|---|---|---|
| 1. C | 2. C | 3. A | 4. BD | 5. ABD |
| 6. CD | 7. ABC | 8. AC | 9. AC | 10. ABC |
| 11. 1 | 12. 4 | 13. 3 | 14. 45 | 15. 7 |
| 16. 0 | 17. 0 | 18. 1.33 | 19. 2 | 20. A-PR,B-PS, C-QS,D-QS |

# GR-04 : TANGENTS AND NORMALS, MONOTONICITY

## SINGLE CORRECT TYPE QUESTIONS

**Q.(1)** Consider the following statements:

S-1: A closed vessel tapers to a point both at its top E and its bottom F and is fixed with EF vertical. When the depth of the liquid in it is $x$ cm, the volume of the liquid in it is $x^2(15-x)$ cu. cm.

S-2: A horse runs along a circle with a speed of 20 km/hr. A lantern is at the center of the circle. A fence is along the tangent to the circle at the point at which the horse starts. The speed with which the shadow of the horse moves along the fence at the moment when it covers $\left(\frac{1}{8}\right)^{th}$ of the circle is $v$ km/hr.

Then the numerical value of $v + EF$ is equal to

(A) 30 (B) 40 (C) 50 (D) 60

**Q.(2)** Let θ be the angle of intersection of curves $\mathbb{C}$ and $\mathbb{D}$. Let ϕ be acute the angle of intersection of curves $\mathbb{E}$ and $\mathbb{F}$. If

$$\mathbb{C}: \frac{x^2}{a^2} + \frac{y^2}{4} = 1, \quad \mathbb{D}: y^2 = 16x, \quad \mathbb{E}: y = |x^2 - 1|, \quad \mathbb{F}: y = |x^2 - 3|,$$

and $\theta = \frac{\pi}{2}$, then value of $a^2 + 7\tan\theta$ is equal to

(A) $2 + 4\sqrt{2}$ (B) $4\sqrt{2} - 2$ (C) $2 + 3\sqrt{2}$ (D) $3 + 4\sqrt{2}$

**Q.(3)** Let

$$\mathbb{A} = \left\{k \in \mathbb{R} \,\middle|\, \alpha, \beta, \gamma \text{ are roots of } \frac{x^3}{a^2} + \left(a^2 + 4 + \frac{1}{a^2}\right)x = \frac{x^2}{a^2} + 1\right\}$$

$$\mathbb{B} = \left\{\left|\frac{a}{2} - \frac{b}{2}\right| \,\middle|\, |a-1| + |b-1| = |a| + |b| = |a+1| + |b+1|, a, b \in \mathbb{R}\right\},$$

$$\lambda = \min(\mathbb{A}), \quad \mu = \min(\mathbb{B}), \quad k = \sum_{\alpha,\beta,\gamma}\left(\frac{\alpha}{\beta}+\frac{\beta}{\alpha}\right).$$

Then which of the following options is/are equal to $\lambda + \mu$?

(A) 5 (B) 2 (C) 3 (D) 4

**Q.(4)**

Let $f : R - \{a, b\} \to R$, $f(x) = \dfrac{(x-1)(x-3)}{(x-a)(x-b)}$, then which of the following is / are correct :-

(A) $f(x) = K$ has 3 distinct solution $\forall\ K \in (0, \infty)$ and $\forall\ a, b \in (3, \infty)$ and $(a < b)$
(B) $f(x) = K$ has two solutions distinct $\forall\ K \in (0, \infty)$ if $a \in (1, 3)$ and $b \in (3, \infty)$
(C) range of $f(x)$ can-not be R if $a, b \in (3, \infty)$ and $(a < b)$
(D) $f(x)$ is strictly increasing in the interval $(-\infty, a)$ if $a, b \in (3, \infty)$ and $(a < b)$

## MULTIPLE CORRECT TYPE QUESTIONS

**Q.(5)** Consider the following statements:

S-1: A point $P(x, y)$moves on curve $\mathbb{C}: x^{\frac{2}{3}} + y^{\frac{2}{3}} = k^{\frac{2}{3}}; k > 0$, such that perpendiculars drawn from origin on tangent and normal at P, are of length $p_1(x)$ and $p_2(x)$ respectively. Let

$$\lambda = \frac{dp_1}{dx}\cdot\frac{dp_2}{dx},$$

S-2: Sides of $\Delta$ABC vary in such a way that its circumradius remains constant. Let

$$\mu = \frac{da}{\cos A} + \frac{db}{\cos B} + \frac{dc}{\cos C},$$

where a,b,c represent the sides BA,CA,AB of $\Delta$ABC (dx represents small increment in variable x).

Then which of the following options is/are correct?

(A) $\lambda + \mu \geq 0$ (B) $\lambda + \mu \leq 0$ (C) $\lambda + \mu = 0$ (D) $\lambda \leq 0$

**Q.(6)** Let L denote a tangent of $y^2 - 2x^3 - 4y + 8 = 0$ passing through the point (1,2). Also let P be the point of intersection of tangents drawn to $x^2 y = 1 - y$ at points where it is intersected by the curve $xy = 1 - y$. Then which of the following options is/are correct?

(A) $L: 2\sqrt{3}x - y = 2(\sqrt{3} - 1)$ (B) $P = \left(0, \frac{1}{2}\right)$

(C) $2\sqrt{3}x + y = 2(\sqrt{3} + 1)$ (D) $P = (0,1)$

**Q.(7)** Which of the following options is/are correct?

(A) 1 is the length of x-intercept of tangent to both branches of function

$$f(x) = \begin{cases} -x^2, & x < 0 \\ x^2 + 8, & x \geq 0 \end{cases}.$$

(B) If the area of triangle formed by tangent to $xy^n = a^{n+1}$and coordinate axes is a constant, then n=1,

(C) The segment of normal to curve $x = 2a \sin t + a \sin t \cos^2 t\,;\, y = -a \cos^3 t$ contained between coordinate-axes is equal to 2a.

(D) The area of triangle formed by tangent to $xy^n = a^{n+1}$ and coordinate axes is constant, for infinitely many real values of n.

**Q.(8)** Consider the following curves, where t is a parameter

$$\mathbb{C}: x = \sqrt{a} \sec t\,, y = \sqrt{b} \csc t\,, \quad \mathbb{D}: x = \sec^2 t\,, y = \cot t.$$

If λ is the x-intercept of the tangent of ℂ at any arbitrary point $P\ (\alpha, \beta)$, and tangent to 𝔻 at $A\left(t = \frac{\pi}{4}\right)$ meets it again at B, then

(A) $\lambda$ is proportional to $\alpha^3$ (B) $2AB = 3\sqrt{5}$

(C) $\lambda$ is proportional to $\sqrt[3]{\alpha}$ (D) $3AB = 2\sqrt{5}$

**Q.(9)** Let $f: \mathbb{R} \to \mathbb{R}$ be a differentiable function satisfying

$$f'(x) > 2\, f(x) \forall x \in \mathbb{R}; f(0) = 1.$$

Then which of the following options is/are correct for all $x \in (0, \infty)$?

(A) $f(x) > e^{2x}$ (B) $f(x) < -e^{2x}$ (C) $|f(x)| > e^{2x}$ (D) $f'(x) > 0$

**Q.(10)** If $ax + by + c = 0$ is equation of a line which is tangent at one point and normal at another point of the curve $27\,y^2 = 4\,x^3$, then which of the following option(s) is/are possible?

(A) $c = 2a$ (B) $c + 2a = 0$ (C) $a = b\sqrt{2}$ (D) $a = -b\sqrt{2}$

**Q.(11)** Let tangent to $\mathbb{C}$ at $P(1,7)$ touches the circle $\mathbb{D}$ at $(\alpha, \beta)$, where

$$\mathbb{C}: y = x^2 + 6, \quad \mathbb{D}: x^2 + y^2 + 16x + 12y + c = 0, \quad \mathbb{E}: y = e^x.$$

If $A(c, e^c)$, $B(c-1, e^{c-1})$, $C(c+1, e^{c+1})$, then which of the following options is/are correct?

(A) Tangent of $\mathbb{E}$ at A intersects AB on LHS of A
(B) $2\alpha - 3\beta = 9$
(C) Tangent of $\mathbb{E}$ at A intersects AB on RHS of A
(D) $3\alpha - 2\beta = 4$

**Q.(12)** Let $|\mathbb{X}|$denotes the cardinal number of set $\mathbb{X}$,

$$\mathbb{A} = \{x \in \mathbb{R} \mid x^4 - 4\,x^3 + 12\,x^2 + x - 1 = 0\},$$

$$\mathbb{B} = \{x \in \mathbb{R} \mid x^5 - 5x + a = 0\}.$$

Then, which of the following options is/are correct?

(A) $a > 2|\mathbb{A}| \Longrightarrow |\mathbb{B}| = 1$
(B) $|\mathbb{B}| = 1 \Longrightarrow |a| > 2|\mathbb{A}|$
(C) $|a| < 2|\mathbb{A}| \Longrightarrow |\mathbb{B}| = 3$
(D) $|\mathbb{B}| = 3 \Longrightarrow |a| \le 2|\mathbb{A}|$

**Q.(13)** Let function $f$ be defined as

$$f: \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) \to \mathbb{R};\, f(x) = (\ln(\sec x + \tan x))^3.$$

Then which of the following options is/are correct?

(A) $f$ is an even function
(B) $f$ is an odd function
(C) $f$ is an injective function
(D) $f$ is a surjective function

**Q.(14)** Which of the following options is/are correct?

(A) $x \in (0,1) \Longrightarrow \sin \pi x \le 4x(1-x)$

(B) $f(x) = -\sin^3 x + 3\sin^2 x + 5$ is increasing in $x \in \left(0, \frac{\pi}{2}\right)$

(C) $f(x) = -\sin^3 x + 3\sin^2 x + 5$ is decreasing in $x \in \left(-\frac{\pi}{2}, 0\right)$

(D) $f(x) = \sec x - \csc x$ is increasing in $x \in \left(0, \frac{\pi}{2}\right)$

## NUMERICAL ANSWER TYPE QUESTIONS

**Q.(15)** Let $P\ (x_1,\ y_1)$ be a point on curve

$$\mathbb{C}: (x^2 - 11)(y + 1) + 4 = 0,$$

where $x_1, y_1 \in \mathbb{N}$. If the area of the triangle formed by normal drawn to $\mathbb{C}$ at P and the coordinate axes is equal to

$$\frac{a}{b}; a, b \in \mathbb{N} \quad \text{and} \quad \lambda = \min(a - 6b),$$

then find the value of λ.

**Q.(16)** Find λ, if

$$\left\{a \in \mathbb{R} \,\middle|\, \{x \in \mathbb{R}^- \mid 3 - x^2 > |x - a|\} \text{ is NOT a NULL set}\right\} = \left(-\frac{13}{4}, \lambda\right).$$

**Q.(17)** Let function $f$ be defined as

$$f: \mathbb{R} \to \mathbb{R}; f(x) = ax + \cos 2x + \sin x + \cos x\,.$$

Further $m, n \in \mathbb{Z}$ such that

$$\left\{a \in \mathbb{R} \,\middle|\, f \text{ is a strictly increasing function.}\right\} = \left[\frac{m}{n}, \infty\right); \gcd(m, n) = 1.$$

Then the value of $(m - n)$ is equal to

**Q.(18)**

Let $p$ be a 6-degree polynomial satisfying

$$p(x) = p(2 - x) \forall x \in \mathbb{R}, \quad \text{and} \quad |\{x \in \mathbb{R} \mid p(x) = 0\}| = 5.$$

If S denotes the sum of roots of polynomial $p$, and $a$ is a positive constant satisfying

$$\{x \in \mathbb{R} \mid f''(x) > 0\} = \left(-\frac{\lambda}{a}, \frac{a}{\mu}\right), \quad f(x) = \begin{cases} xe^{ax}, & x \le 0 \\ x + ax^2 - x^3, & x > 0 \end{cases},$$

then $\lambda + \mu + S$ is equal to (Here $|\mathbb{X}|$ denotes the cardinal number of set $\mathbb{X}$ )

**Q.(19)** Let $r_f$ denote the number of distinct real roots of a polynomial $f$. Suppose

$$\mathbb{A} = \left\{(x^2-1)^2(ax^3+bx^2+cx+d) \;\middle|\; a,b,c,d \in \mathbb{R}\right\}.$$

If $f \in \mathbb{A}$, then minimum possible value of $r_{f'} + r_{f''}$ is equal to

**Q.(20)** Let $\theta$ be acute angle between $y = x^x \ln x$ and $y \ln 2 = 2^x - 2$ at their point of intersection on line $y = 0$. If

$$\left\{a \in \mathbb{R} \middle| f(x) = \sin x - a\sin 2x - \frac{1}{3}\sin 3x + 2ax; f'(x) \ge 0\ \forall x \in \mathbb{R}\right\} = [\lambda, \infty),$$

then $7\lambda + 9\tan\theta$ is equal to

## PASSAGES

**Passage-I:**

A function $f: \mathbb{R} \to \mathbb{R}$ satisfies the following properties:

$f'(x) > 0\ \forall x \in (-\infty, -2) \cup (3, \infty)$, $\quad f'(x) \le 0\ \forall x \in (-2,3)$, $\quad \lim_{x\to\infty} f(x) = 0$,

$f$ is continuous $\forall$ x $\in \mathbb{R} - \{3\}$, $\quad f$ is differentiable $\forall$ x $\in \mathbb{R} - \{-2,3\}$, $\quad \lim_{x\to 3} f(x) \to -\infty$,

$f''(x) > 0 \forall x \in (-\infty, 0) - \{-2\}$, $\quad f''(x) < 0\ \forall\ x \in (0,\infty) - \{3\}$, $\quad \lim_{x\to -\infty} f(x) = 3$,

and $f(0) = 0$. Now, answer the following questions:

**Q.(21)** Let λ be the maximum number of solutions of $f(x) = |x|$. If

$$f'(0) > -3 \text{ and } f(-2) > 6,$$

then number of solutions of $f(x) + 3x = 0$ is equal to

(A) $\lambda + 1$ (B) $\lambda - 1$ (C) $2\lambda - 1$ (D) $3\lambda - 4$

**Q.(22)** The function $y = f(-|x|)$ is

(A) differentiable $\forall x \in \mathbb{R}$, if $f'(0) = 0$.

(B) continuous but NOT differentiable at two points, if $f'(0) = 0$.

(C) continuous but NOT differentiable at one point, if $f'(0) = 0$.

(D) dis-continuous at two points if $f'(0) = 0$.

**Passage-II:**

Let $|\mathbb{X}|$ denotes the cardinal number of set $\mathbb{X}$,

$$\mathbb{A} = \{27\lambda \in \mathbb{R} \mid |\mathbb{B}| = 3\}, \mathbb{B} = \{x \in \mathbb{R} \mid x^3 + x^2 - x + \lambda = 0\} \text{ and } f(x) = 2 - \sqrt{4 - (x-1)^2}.$$

Now, answer the following questions:

**Q.(23)** Which of the following options lie in set $\mathbb{A}$?

(A) 5 (B) $-16$ (C) $-27$ (D) 4

**Q.(24)** Which of the following options is/are correct?

(A) $f$ is monotonic in interval $(-1,1)$.

(B) $f$ is monotonic in interval $(1,3)$.

(C) $y = 2x - 2\sqrt{5}$ is tangent to $f$

(D) $y = -3x + 5 + 2\sqrt{10}$ is tangent to $f$

## MATRIX MATCHING

**Q.(25)** Match the curves in Column-I with curves in Column-II, if they touch each other.

| Column-I | Column-II |
|---|---|
| (A) $y^2 = 4x$ | (P) $x^2 + 676y^2 = 676$. |
| (B) $xy = 4$ | (Q) $5x - 4y = 1$. |
| (C) $\sqrt[3]{x^2} + \sqrt[3]{y^2} = 9$ | (R) $x^2 + y^2 = 8$. |
| (D) $x = 2t - t^2, y = t + t^2$ | (S) $x^2 + y^2 = 6x - 1$. |

# KEY

| | | | | |
|---|---|---|---|---|
| 1. C | 2. A | 3. D | 4. C | 5. BD |
| 6. ABD | 7. ABC | 8. AB | 9. ACD | 10. BCD |
| 11. AB | 12. ACD | 13. BCD | 14. ABCD | 15. 3 |
| 16. 3 | 17. 9 | 18. 11 | 19. 5 | 20. 10 |
| 21. A | 22. B | 23. BD | 24. ABC | 25. A-S,B-R, C-P,D-Q |

# GR05-MAXIMA-MINIMA, MVT

## SINGLE CORRECT TYPE QUESTIONS

**Q.(1)** Which of the following options is/are correct, if $m_{\mathbb{X}}$ denotes the smallest element in a set $\mathbb{X}$,

$$\mathbb{A}=\{e^{x^2}+e^{-x^2}|x\in[0,1]\},\quad \mathbb{B}=\{xe^{x^2}+e^{-x^2}|x\in[0,1]\},\quad \mathbb{C}=\{x^2e^{x^2}+e^{-x^2}|x\in[0,1]\}?$$

(A) $m_{\mathbb{A}}=m_{\mathbb{B}}\neq m_{\mathbb{C}}$ (B) $m_{\mathbb{B}}\neq m_{\mathbb{A}}=m_{\mathbb{C}}$ (C) $m_{\mathbb{A}}\neq m_{\mathbb{B}}\neq m_{\mathbb{C}}$ (D) $m_{\mathbb{A}}=m_{\mathbb{B}}=m_{\mathbb{C}}$

**Q.(2)**

Let $f(x)=\begin{cases}5, & x=0\\ x^2+5x+\alpha, & 0<x<2\\ \beta x+\gamma, & 2\le x\le 3\end{cases}$

If $f(x)$ satisfies all the conditions of Lagranges mean value theorem in $[0,3]$ and $P(c,f(c))$ is the point on the curve $f(x)$ in $c\in[0,3]$ where the tangent is parallel to the chord joining the end points, then find the value of $\frac{1}{17}(\alpha+2\beta+3\gamma+6c)$

(A) 1 (B) 2 (C) 3 (D) 4

**Q.(3)**

If f(x) is a differentiable function and $0<a_1<a_2<....<a_{100}<100$ and $f(a_r)=(-1)^r.r,\ 1\le r\le 99$ and $f(a_{100})=0$ and roots of equation $f(x)=0$ are rational then the minimum number of zeroes of $f(x).\cos x+f'(x).\sin x$ for $x\in[0,100]$ is

(A) 129 (B) 130 (C) 131 (D) 128

**Q.(4)**

Let $f(x)$ satisfy the requirement of Lagrange's mean value theorem in [0,2]. If $f(0)=0$ and $|f'(x)| \le \frac{1}{2}$ for all $x \in [0,2]$, then which of the following is not true ?

(A) $f(x) \le 2$

(B) $|f(x)| \le 2x$

(C) $f(x) = 3$ for at least one $x \in [0,2]$

(D) $|f(x)| \le 1$

## MULTIPLE CORRECT TYPE QUESTIONS

**Q.(5)**

Let $f : R \to R$ be defined by $f(x) = \begin{cases} 2x^2 \ln|x| - 5x^2, & x \ne 0 \\ 0, & x = 0 \end{cases}$. Which of the following statement(s) is/are correct?

(A) f(x) has exactly one local maximum and two local minimum points

(B) f(x) is strictly increasing in $(10, \infty)$

(C) Absolute minimum value of f(x) exist but absolute maximum value of f(x) does not exist

(D) In the interval $x \in (0, \infty)$, $f(x) = k$ has two distinct roots

**Q.(6)**

Which of the following statement(s) is/are correct ?

(A) If $f(x) = x^2 + 10 \sin x$, $-\infty < x < \infty$ then $\exists$ some $c \in R$ such that $f(c) = 2015$.

(B) Suppose f is continuous function on [1, 5] and the only solutions of the equation $f(x) = 6$ are $x = 1$ and $x = 4$. If $f(2) = 8$, then $f(3) > 6$.

(C) The functions $f(x) = \text{sgn}(\tan^{-1} e^{-x})$ and $g(x) = \cos\left(2[\cot^{-1} x]\pi\right)$ are identical functions, where sgn (k) and [k] denote signum function of k and greatest integer less than or equal to k respectively.

(D) If the function $g(x) = \text{sgn}(kx)$ is continuous for every $x \in R$, then number of real values of k is one. (where sgn(x) denotes signum function of x).

**Q.(7)**

Let $f(x)=\dfrac{(x-1)^2.e^x}{(1+x^2)^2}$, then which of the following statement(s) is(are) correct?

(A) f(x) is strictly increasing in $(-\infty, -1)$

(B) f(x) is strictly decreasing in $(1, \infty)$

(C) f(x) has two points of local extremum

(D) f(x) has a point of local minimum at some $x \in (-1, 0)$

**Q.(8)**

If f(x) is double differentiable function such that $|f''(x)| \le 5$ for each x in the interval [0, 4] and f takes its largest value at an interior point of this interval, then value of $|f'(0)| + |f'(4)|$ cannot be :-

(A) 10 (B) 15 (C) 21 (D) 28

**Q.(9)** Which of the following options is/are correct?

(A) If $f$ and $g$ are continuous functions in $[a, b]$ and differentiable in $(a, b)$ then there exists at least one $c \in (a, b)$ such that $\begin{vmatrix} f(a) & f(b) \\ g(a) & g(b) \end{vmatrix} = (b-a)\begin{vmatrix} f(a) & f'(c) \\ g(a) & g'(c) \end{vmatrix}$.

(B) If functions $f, g, h$ are continuous in $[a, b]$ and differentiable in $(a, b)$, then there exists at least one $c \in (a, b)$ such that $\begin{vmatrix} f(a) & f(b) & f'(c) \\ g(a) & g(b) & g'(c) \\ h(a) & h(b) & h'(c) \end{vmatrix} = 0$.

(C) Let $a > 0$ and $f$ be continuous in $[-a, a]$ and $f'(x) \le 1 \forall x \in (-a, a)$. If $f(a) = a, f(-a) = -a$, then f(0)=0.

(D) Let $f$ is continuous in [a,b] and differentiable in (a,b). If $f(a) = a$ and $f(b) = b$, then there exists $(c_1, c_2) \subseteq (a, b)$ such that $f'(c_1) + f'(c_2) = 2$.

**Q.(10)** Let $x = a$ be a point where $f(x) = 2|x| + |x+2| - \big||2|x| - |x+2|\big|$ has a local extremum, and

$$\beta = \min\left\{k \in \mathbb{R} \,\middle|\, 4k\,x^2 + \frac{1}{x} \ge 1 \forall x > 0\right\}.$$

Then which of the following option(s) is/are possible?

(A) $54\beta = -a$ (B) $81\beta = -a$ (C) $54\beta = -3a$ (D) $27\beta = -2a$

**Q.(11)** Let $f: \mathbb{R} \to \mathbb{R}^+$ and $g: \mathbb{R} \to \mathbb{R}$ be twice differentiable functions such that $f''$ and $g''$ are continuous functions on $\mathbb{R}$. Suppose $(f'(2), g(2)) = (0,0) \neq \left(f''(2), g'(2)\right)$,

$$\lim_{x \to 2} \frac{f(x)g(x)}{f'(x)g'(x)} = 1 \text{ and } h(x) = \begin{vmatrix} \cos 2x & \cos 2x & \sin 2x \\ -\cos x & \cos x & -\sin x \\ \sin x & \sin x & \cos x \end{vmatrix}.$$

Then which of the following options is/are INCORRECT?

(A) $f$ has local minimum at $x = 2$, and $h'$ has exactly three solutions in $(-\pi, \pi)$

(B) $f$ has local maximum at $x = 2$, and $h$ attains maximum at $x = 0$.

(C) $f''(2) > f(2)$, and $h$ attains minimum at $x = 0$.

(D) $f(x) = f''(x)$ has at least one real root, and $h'$has more than 3 roots in $(-\pi, \pi)$

**Q.(12)** Which of the following options is/are correct?

(A) There exists $a < c < b$ satisfying

$$\frac{\sin b - \sin a}{\cos a - \cos b} = \cot c \ \forall (a, b) \in \{(x, y) \in \mathbb{R} \times \mathbb{R} | \cos x \neq \cos y\}.$$

(B) For all $0 < a < b$, we have

$$\frac{b - a}{1 + b^2} < \tan^{-1} b - \tan^{-1} a < \frac{b - a}{1 + a^2}.$$

(C) Let $f$ is continuous in interval [a,b] and differentiable in interval (a,b). Then $\forall (a, b) \subseteq \mathbb{R} - \{0\}$, there exists at least one $c \in (a, b)$ for which

$$\frac{f(b) - f(a)}{\frac{1}{b} - \frac{1}{a}} = -c^2 f(c).$$

(D) If $f$ is differentiable in interval $[1,5]$, then there exists $(a, b) \subseteq [1,5]$ satisfying

$$\frac{f^2(5) - f^2(1)}{f'(a)f(b)} = 8.$$

**Q.(13)** For every twice differentiable function $f:\mathbb{R}\to[-2,2]$ with $\left(f(0)\right)^2+\left(f'(0)\right)^2=10$, which of the following options is/are correct?

(A) There exists interval $I_1\subseteq\mathbb{R}$, and $I_2\subseteq\mathbb{R}$ such that $f$ is one-one in $I_1$ and many-one in $I_2$.

(B) There exists $\alpha\in(-4,0)$ and $\beta\in(0,4)$ such that $|f'(\alpha)|\le 1$ and $|f'(\beta)|\le 1$.

(C) $f$ must be a many-one function satisfying $\lim\limits_{n\to\infty} f(n)=1$.

(D) There exists $\alpha\in(-4,4)$ such that $f(\alpha)+f''(\alpha)=0$ and $f'(\alpha)\neq 0$

**Q.(14)** Which of the following options is/are correct for

$$f(x)=\begin{cases}\sqrt{x}\ln x, & x>0\\ 0, & x=0\end{cases}?$$

(A) $f$ is continuous at $x=0$ and has no absolute maxima.

(B) Smallest possible value of $f$ is $-\frac{1}{e}$.

(C) $f'(0)=d.n.e$ and $\lim\limits_{x\to 0} f'(x)=d.n.e.$

(D) $f$ has an inflection point at $x=1$.

## NUMERICAL ANSWER TYPE QUESTIONS

**Q.(15)** A conical vessel is to be prepared out of a circular sheet of gold of unit radius. If $\lambda\pi$ is the sectorial area which is used to form the sheet so that the vessel has maximum volume, then the integer nearest to value of $\sqrt{300}\lambda$ is equal to

**Q.(16)** Find $13(\max\mathbb{A})+2(\min\mathbb{B})$, if

$$\mathbb{A}=\left\{\frac{1}{|x-4|+1}+\frac{1}{|x-8|+1}\ \middle|\ x\in\mathbb{R}\right\},\quad \text{and}\quad \mathbb{B}=\left\{\frac{b+1}{a+b-2}\in\mathbb{R}\ \middle|\ a^2+b^2=1\right\}.$$

**Q.(17)**

Let A (1, –1), B (4, –2) and C (9, 3) be the vertices of the triangle ABC. A parallelogram AFDE is drawn with vertices D, E and F on the line segments BC, CA and AB respectively. Find the maximum area of parallelogram AFDE.

**Q.(18)** From a fixed point A on circumference of a circle of radius $\sqrt{2\sqrt{3}}$, let the perpendicular AB fall on the tangent at C on it. Then greatest possible area of $\Delta$ABC is equal to

**Q.(19)** Consider all the rectangles lying in the region

$$\left\{(x,y) \in \mathbb{R}^2 \,\middle|\, 0 \le x \le \frac{\pi}{2} \text{ and } 0 \le y \le 2\sin 2x\right\}$$

and having one side on X-axis. The area of the rectangle which has maximum perimeter among all such rectangles, is $k\pi$. Then value of $k\sqrt{3}$ is equal to

**Q.(20)** Let $f:\mathbb{R} \to \mathbb{R}$ and $g:\mathbb{R} \to (0,\infty)$ are functions satisfying

$$f(x) = \ln g(x), \quad f'(x) = 2010(x-2009)(x-2010)^2(x-2011)^3(x-2012)^4.$$

Then number of points of local minima of function $g$ is/are

## PASSAGES

**Passage-I:**

Let $f:\mathbb{R}^+ \to \mathbb{R}$ and $g:\mathbb{R}^+ \to \mathbb{R}$ be functions defined by

$$f(x) = \int_0^x \prod_{r=1}^{21} (t-r)^r \, dt\,; x > 0 \qquad g(x) = 98(x-1)^{50} - 600(x-1)^{49} + 2450; x > 0.$$

Let $m_p$ and $M_p$ denote the number of points of local minimum and local maximum of a certain function $p$. Then, answer the following questions:

**Q.(21)** The value of $2m_f + 3M_f + m_f M_f$ is equal to

**Q.(22)** The value of $6m_g + 4M_g + 8m_g M_g$ is equal to

**Passage-II:**

If $f(x)$ is defined in [a,b] and (i) $f(x)$ is continuous on [a,b]
(ii) $f(x)$ is derivable on (a,b)
(iii) $f(a)=f(b)$. Then has atleast one point $c \in (a,b)$ such that
$f'(c)=0$. This is known as Rolle's theorem. This means $f(x)$ is a polynomial then between any two roots of $f(x)=0$ there is always lies a root of $f'(x)=0$

**Q.(23)**

A twice differentiable function f such that $f(a)=f(b)=0$ and $f(c)>0$, $a<c<b$. Then there is atleast one value 'ε' between a and b for which

(A) $f''(\varepsilon)=0$ (B) $f''(\varepsilon)>0$ (C) $f''(\varepsilon)<0$ (D) $f''(\varepsilon)\geq 0$

**Q.(24)** If $f$ is continuous in [a,b] and differentiable in (a,b), then

$$\frac{f'(c)}{3c^2}=\frac{f(b)-f(a)}{b^3-a^3}$$

holds for

(A) at least one $c \in (a,b)$ (B) at most one $c \in (a,b)$

(C) exactly one $c \in (a,b)$ (D) no $c \in (a,b)$

## MATRIX MATCHING

**Q.(25)**

Let f(x) be a real valued function defined by $f(x)=x^2-2|x|$ and

$$g(x)=\begin{cases}\text{minimum}\{f(t);-2\leq t\leq x\}, & x\in[-2,0)\\ \text{maximum}\{f(t);0\leq t\leq x\}, & x\in[0,3]\end{cases}$$

Now, match the following columns:

| Column-I | Column-II |
|---|---|
| (A) $g$ is NOT continuous at $x=$ | (P) $-2$ |
| (B) $g$ is NOT differentiable at $x=$ | (Q) 0 |

(C) Number of points of strict local extremum of $g$ is (R) 1

(D) Maximum value of $g$ is equal to (S) 3

## KEY

| | | | | |
|---|---|---|---|---|
| 1. D | 2. B | 3. B | 4. C | 5. ABC |
| 6. ABCD | 7. AC | 8. CD | 9. ABCD | 10. AC |
| 11. AC | 12. ABD | 13. ABD | 14. ACD | 15. 3 |
| 16. 32 | 17. 5 | 18. 2.25 | 19. 0.5 | 20. 1 |
| 21. 57 | 22. 6 | 23. C | 24. A | 25. A-Q,B-Q, C-Q,D-S |

# GR-06 : INTEGRATION

## SINGLE CORRECT TYPE QUESTIONS

**Q.(1)**

Let f(t) be a continous function for $t \in R$. Let $I_1 = \int_{\sin^2 t}^{1+\cos^2 t} x f\left(x(2-x)\right)dx$ and $I_2 = \int_{\sin^2 t}^{1+\cos^2 t} f\left(x(2-x)\right)dx$,

then :-

(A) $\frac{I_1}{I_2} = 2$ (B) $\frac{I_1}{I_2} = 1$ (C) $\frac{I_1}{I_2}$ is not difined (D) $\frac{I_1}{I_2} = \frac{1}{2}$

**Q.(2)**

The absolute value of $\dfrac{\int_0^{\pi/2}\left(x\cos x+1\right)e^{\sin x}dx}{\int_0^{\pi/2}\left(x\sin x-1\right)e^{\cos x}dx}$ is equal to -

(A) e (B) $\pi e$ (C) e/2 (D) $\pi/e$

**Q.(3)**

The value of definite integral $\int_{-\pi}^{\pi}\frac{x^2}{1+\sin x+\sqrt{1+\sin^2 x}}dx$, is :

(A) $\frac{3x^2}{2}$ (B) $\frac{\pi^3}{3}$ (C) $\frac{2\pi^3}{3}$ (D) $\frac{\pi^3}{6}$

**Q.(4)** Which of the following option(s) is/are equal to

$$I = \int \frac{\cos 2x \sin 4x}{\cos^4 x\,(1+\cos^2 2x)}\,dx\,.$$

Here C denotes the constant of integration.

(A) $\sec^2 x - \ln(1+\cos^2 2x) + 2\ln(1+\cos 2x) + C$

(B) $\sec^2 x + \ln(1+\cos^2 2x) + 2\ln(1+\cos 2x) + C$

(C) $\sec^2 x - \ln(1+\cos^2 2x) - 2\ln(1+\cos 2x) + C$

(D) $\sec^2 x + \ln(1+\cos^2 2x) - 2\ln(1+\cos 2x) + C$

**Q.(5)** Let

$$I = \int \frac{e^x}{e^{4x}+e^{2x}+1}dx, \quad J = \int \frac{e^{-x}}{e^{-4x}+e^{-2x}+1}dx,$$

$$K = \frac{1}{2}\ln\frac{e^{4x}-e^{2x}+1}{e^{4x}+e^{2x}+1} \quad \text{and} \quad L = \frac{1}{2}\ln\frac{e^{2x}-e^{x}+1}{e^{2x}+e^{x}+1}.$$

Then which of the following options is/are correct? (Here C denotes the constant of integration.)

(A) $I - J = K + C$ (B) $I - J = C - L$ (C) $I + J = C - K$ (D) $I + J = L + C$

## MULTIPLE CORRECT TYPE QUESTIONS

**Q.(6)**

Let $I_1 = \int_0^{\pi/2} \cos(\pi \sin^2 x)dx$, $I_2 = \int_0^{\pi/2} \cos(2\pi \sin^2 x)dx$ and $I_3 = \int_0^{\pi/2} \cos(\pi \sin x)dx$ then -

(A) $I_1 = 0$ (B) $I_2 + I_3 = 0$ (C) $I_1 + I_2 + I_3 = 0$ (D) $I_2 + I_3 \neq 0$

**Q.(7)**

Let $f(x)$ be a non-constant twice differentiable function defined on $(-\infty, \infty)$ such that $f(x) = f(1-x)$ and $f'(1/4) = 0$. Then

(A) $f''(x)$ vanishes at least twice on $[0, 1]$ (B) $f'(1/2) = 0$

(C) $\int_{-1/2}^{1/2} f\left(x+\frac{1}{2}\right)\sin x\,dx = 0$ (D) $\int_0^{1/2} f(t)e^{\sin \pi t}dt = \int_{1/2}^{1} f(1-t)e^{\sin \pi t}dt$

**Q.(8)** Suppose $a$ and $b$ are positive real numbers such that $ab = 1$. Let for any real parameter $t$, the distance from the origin to the line $(ae^t)x + (be^{-t})y = 1$ be denoted by D(t). If

$$I = \int_0^1 \frac{dx}{(D(x))^2},$$

then which of the following options is/are correct?

(A) The value of I is $\frac{e^2-1}{2}\left(b^2 + \frac{a^2}{e^2}\right)$.

(B) I is minimum when $b = \sqrt{e}$

(C) Minimum value of I is $e - \frac{1}{e}$

(D) The value of I is $\frac{e^2+1}{2}\left(b^2 + \frac{a^2}{e^2}\right)$

**Q.(9)**

If $\int \cos(2016x).\left(\sin^{2014} x\right)dx = f(x) + c, f(0) = 0$ then -

(A) f(x) is odd

(B) $\int_{\pi}^{2\pi} f(x)dx = 0$

(C) $\int_0^{\pi} f(x)dx = 0$

(D) $\lim_{x\to 0} \frac{2015 f(x)}{x^{2015}} = 1$

**Q.(10)** Which of the following option(s) satisfy the equation

$$\int_{-\infty}^{\infty} \frac{\ln|t|}{t^2 + x^4} dx = \frac{\pi}{2}\ln 2\,;\, x \in \mathbb{R}.$$

(A) Number of values of x satisfying the equation is 5

(B) Roots of $x^2 - \left(2+\sqrt{2}\right)x + 2\sqrt{2} = 0$, satisfy the above equation

(C) Product of values of x satisfying above equation is 8

(D) Roots of $x^3 + 31x^2 - 2x - 21 = 0$ satisfy the above equation

**Q.(11)** Choose the correct option(s), if

$$\int_{\frac{3}{2}}^{x} \frac{x^2+3}{x^4+x^3-x^2-3x+9} dx = f(x).$$

(A) $f(2) = \frac{2}{\sqrt{19}}\tan^{-1}\frac{2}{\sqrt{19}}$

(B) $f(2) = \frac{2}{\sqrt{19}}\tan^{-1}\frac{4}{\sqrt{19}}$

(C) $f(3) = \frac{2}{\sqrt{19}}\tan^{-1}\frac{5}{\sqrt{19}}$

(D) $f(3) = \frac{2}{\sqrt{19}}\tan^{-1}\frac{3}{\sqrt{19}}$

**Q.(12)** Let

$$f(x)=\int_1^x \frac{x^2-1}{x^3\cdot\sqrt{2x^4-2x^2+1}}dx.$$

Then which of the following options is/are correct?

(A) $f(2)=\frac{1}{4}$

(B) $f(2)=\frac{1}{8}$

(C) $f(\sqrt{2})=\sin\frac{\pi}{10}$

(D) $f(\sqrt{2})=\sin\frac{\pi}{10}-\frac{1}{2}\sin\frac{\pi}{6}$

**Q.(13)** Which of the following options is/are correct?

A) If $f(x)=\sqrt{x},\ and\ g(x)=x^{\frac{1}{3}}, then$

$$\int_0^2 \left(f(1+x^3)+g(x^2+2x)\right)dx=6$$

B) Let f : (-1, 1)→ $R$be a twice differentiable function such that $2f'(x)+xf''(x)=1\ \forall$ x, then

$$\int_{-1}^1 x^3 f(x)dx=\frac{1}{5}, if\ f'\left(\frac{1}{2}\right)=\frac{1}{2}$$

C) If $f$ is a function satisfying

$$f(x)=x+\int_0^1 (x+t)f(t)\,dt, then \int_0^1 f(x)dx=-6$$

D) If $f(x)=\ lnx$ then

$$\int_1^{e^2}\frac{f(xe^{x+1})}{(x+1)^2+f^2(x^x)}dx>\frac{\pi}{4}$$

## NUMERICAL ANSWER TYPE QUESTIONS

**Q.(14)** Find the value of I, where

$$I=\int_0^{\frac{\pi}{2}}(\cos(2\pi\sin^2 x)-2\cos(\pi\sin^2 x)+\cos(\pi\sin x))dx.$$

**Q.(15)** Evaluate the integral I, where (take π =3.14)

$$I=\int_{0}^{\pi} e^{\cos x}\cdot\cos(\sin x)\,dx.$$

**Q.(16)** Let us define the integral I, as

$$I=\lim_{n\to\infty}\int_{1}^{n}\frac{x^4+2x^2+1}{\left(x^{\frac{113}{25}}+\frac{11}{3}x^{\frac{63}{25}}+11x^{\frac{13}{25}}\right)^5}dx=\frac{p}{q};p,q\in\mathbb{N};HCF(p,q)=1.$$

Number of divisors of 'p' is equal to

**Q.(17)** Find the value of 3I, where I is defined as

$$I=\sum_{r=0}^{10}\int_{r}^{r+4}|x-1-r|\cdot|x-2-r|\cdot|x-3-r|dx$$

**Q.(18)** Find the area bounded by following

$$f(x)=x^3-6x^2+11x-6;y=0;x=-1;x=4.$$

**Q.(19)** Given that

$$\int_{-1}^{1} tan^{-1}\frac{1}{\sqrt{1-x^2}}\times cot^{-1}\frac{x}{\sqrt{1-x^{2|x|}}}dx=k\pi^2,\text{where } k\in R.$$

Then $(1-k)^2$ is equal to

**Q.(20)** Consider the following two integrals

$$A=\int_{0}^{1}\sqrt{x^7(1-x)^9}dx\quad and\ B=\int_{0}^{1}\sqrt{\frac{x^7(1-x)^9}{(x+5)^{20}}}dx.$$

If $|\ln A-\ln B|=a\ln p_1+b\ln p_2+c\ln p_3$, where $p_1<p_2<p_3$ are prime and $a,b,c\in\mathbb{Q}$, then the value of $a-2b+c$ is equal to

## PASSAGES

**Passage-I:**

Consider the following integrals

$$I = \int_1^2 \frac{1}{x-1+\sqrt{x^2-3x+3}}\,dx \quad and \quad J = \int_1^2 \frac{x^2-x+1}{x(2x-1)^2}\,dx$$

Now, answer the following questions:

**Q.(21)** The value of $I - 2J$ is equal to

**Q.(22)** The value ofsgn$(4(I-J)-1)+8+\text{sgn}(2-4(I-J))$ is equal to

**Passage-II:**

Let

$$\int \frac{x^2}{(x^2-1)\sin 2x + 2x\cos 2x}\,dx = f(x) + C; f\left(\frac{\pi}{2}\right) = \frac{1}{2}\ln\frac{\pi}{2}.$$

Now, answer the following questions:

Here C denotes the constant of integration.

**Q.(23)** The integer nearest to value of $e^{-2f(\pi)}$ is equal to

**Q.(24)** The integer nearest to value of $e^{2f\left(\frac{\pi}{4}\right)}$ is equal to

## MATRIX MATCHING

**Q.(25)** Let

$$a = \int_0^1 \frac{dx}{1+(1-x)^{\frac{\pi}{2}}}, \quad b = \int_0^1 \frac{dx}{1+x^4}$$

$$c = \int_0^1 \frac{dx}{\sqrt{4-x^2+x^5}}, \quad d = \int_0^{\frac{1}{2}} \frac{dx}{\sqrt{1-x^{100}}}$$

Match the truth values of entries in Column-I with those in Column-II.

**Column-I** **Column-II**

| | |
|---|---|
| (A) $a > \ln 2$ | (P) $c < \frac{1}{2}$ |
| (B) $a > \frac{\pi}{4}$ | (Q) $c < \frac{\pi}{6}$ |
| (C) $b < \frac{3}{4}$ | (R) $d > 0.5$ |
| (D) $b < \frac{9}{10}$ | (S) $d > 0.524$ |

## KEY

| | | | | |
|---|---|---|---|---|
| 1. B | 2. A | 3. B | 4. A | 5. B |
| 6. ABC | 7. ABCD | 8. BC | 9. ABCD | 10. BC |
| 11. AC | 12. BD | 13. ABD | 14. 0 | 15. 3.14 |
| 16. 10 | 17. 165 | 18. 18.75 | 19. 0.5 | 20. 1 |
| 21. 0 | 22. 10 | 23. 3 | 24. 8 | 25. A-QR,B-PS,C-PS,D-QR |

# GR-07 : AREA AND DIFFERNTIAL EQUATIONS

## SINGLE CORRECT TYPE QUESTIONS

**Q.(1)**

Let $f:\left[\frac{1}{2},1\right]\to R$ (the set of all real numbers) be a positive, non-constant and differentiable function such that $f'(x) < 2f(x)$ and $f\left(\frac{1}{2}\right)=1$. Then the value of $\int_{1/2}^{1} f(x)dx$ lies in the interval

(A) $(2e - 1, 2e)$ (B) $(e - 1, 2e - 1)$ (C) $\left(\frac{e-1}{2}, e-1\right)$ (D) $\left(0, \frac{e-1}{2}\right)$

**Q.(2)**

Let the functions $f : \mathbb{R} \to \mathbb{R}$ and $g : \mathbb{R} \to \mathbb{R}$ be defined by

$f(x) = e^{x-1} - e^{-|x-1|}$ and $g(x)=\frac{1}{2}\left(e^{x-1}+e^{1-x}\right)$.

Then the area of the region in the first quadrant bounded by the curves $y = f(x)$, $y = g(x)$ and $x = 0$ is

(A) $\left(2-\sqrt{3}\right)+\frac{1}{2}(e-e^{-1})$ (B) $\left(2+\sqrt{3}\right)+\frac{1}{2}(e-e^{-1})$

(C) $\left(2-\sqrt{3}\right)+\frac{1}{2}(e+e^{-1})$ (D) $\left(2+\sqrt{3}\right)+\frac{1}{2}(e+e^{-1})$

**Q.(3)**

The area of the region $\left\{(x, y) : 0 \le x \le \frac{9}{4}, \quad 0 \le y \le 1, \quad x \ge 3y, \quad x + y \ge 2\right\}$ is

(A) $\frac{11}{32}$ (B) $\frac{35}{96}$ (C) $\frac{37}{96}$ (D) $\frac{13}{32}$

**Q.(4)**

A curve passes through the point $\left(1, \frac{\pi}{6}\right)$. Let the slope of the curve at each point (x, y) be $\frac{y}{x} + \sec\left(\frac{y}{x}\right), x > 0$. Then the equation of the curve is

(A) $\sin\left(\frac{y}{x}\right) = \log x + \frac{1}{2}$

(B) $\operatorname{cosec}\left(\frac{y}{x}\right) = \log x + 2$

(C) $\sec\left(\frac{2y}{x}\right) = \log x + 2$

(D) $\cos\left(\frac{2y}{x}\right) = \log x + \frac{1}{2}$

**Q.(5)**

The function $y = f(x)$ is the solution of the differential equation $\frac{dy}{dx} + \frac{xy}{x^2 - 1} = \frac{x^4 + 2x}{\sqrt{1 - x^2}}$ in $(-1, 1)$ satisfying $f(0) = 0$. Then $\int_{-\frac{\sqrt{3}}{2}}^{\frac{\sqrt{3}}{2}} f(x)dx$ is

(A) $\frac{\pi}{3} - \frac{\sqrt{3}}{2}$ (B) $\frac{\pi}{3} - \frac{\sqrt{3}}{4}$ (C) $\frac{\pi}{6} - \frac{\sqrt{3}}{4}$ (D) $\frac{\pi}{6} - \frac{\sqrt{3}}{2}$

**Q.(6)** A tank contains 1000 $\ell$ of water. Brine that contains 0.05 kg/$\ell$ of salt enters the tank at the rate of 5 $\ell$/min. Brine that contains 0.04 kg/$\ell$ of salt enters the tank at the rate of 10 $\ell$/min. The solution is kept thoroughly mixed and drains from the tank at the rate of 15 $\ell$/min. If the amount of salt left in the tank after 200 minutes, is A, then 3A is equal to

(A) $130(1-e^{-1})$ (B) $130(1-e^{-2})$ (C) $130(1-e^{-3})$ (D) $130(1-3e^{-1})$

## MULTIPLE CORRECT TYPE QUESTIONS

**Q.(7)**

Let $\Gamma$ denote a curve $y = y(x)$ which is in the first quadrant and let the point $(1, 0)$ lie on it. Let the tangent to $\Gamma$ at a point P intersect the y-axis at $Y_P$. If $PY_P$ has length 1 for each point P on $\Gamma$, then which of the following options is/are correct ?

(1) $y = \log_e\left(\frac{1+\sqrt{1-x^2}}{x}\right) - \sqrt{1-x^2}$

(2) $xy' - \sqrt{1-x^2} = 0$

(3) $y = -\log_e\left(\frac{1+\sqrt{1-x^2}}{x}\right) + \sqrt{1-x^2}$

(4) $xy' + \sqrt{1-x^2} = 0$

**Q.(8)**

Let b be a nonzero real number. Suppose $f : \mathbb{R} \to \mathbb{R}$ is a differentiable function such that $f(0)=1$. If the derivative $f'$ of $f$ satisfies the equation $f'(x) = \frac{f(x)}{b^2 + x^2}$

for all $x \in \mathbb{R}$, then which of the following statements is/are TRUE ?

(A) If $b > 0$, then $f$ is an increasing function

(B) If $b < 0$, then $f$ is a decreasing function

(C) $f(x)\, f(-x) = 1$ for all $x \in \mathbb{R}$

(D) $f(x) - f(-x) = 0$ for all $x \in \mathbb{R}$

**Q.(9)**

For any real numbers $\alpha$ and $\beta$, let $y_{\alpha,\beta}(x)$, $x \in \mathbb{R}$, be the solution of the differential equation

$$\frac{dy}{dx} + \alpha y = xe^{\beta x},\ y(1) = 1$$

Let $S = \{y_{\alpha,\beta}(x) : \alpha, \beta \in \mathbb{R}\}$. Then which of the following functions belong(s) to the set S ?

(A) $f(x) = \frac{x^2}{2}e^{-x} + \left(e - \frac{1}{2}\right)e^{-x}$

(B) $f(x) = -\frac{x^2}{2}e^{-x} + \left(e + \frac{1}{2}\right)e^{-x}$

(C) $f(x) = \frac{e^x}{2}\left(x - \frac{1}{2}\right) + \left(e - \frac{e^2}{4}\right)e^{-x}$

(D) $f(x) = \frac{e^x}{2}\left(\frac{1}{2} - x\right) + \left(e + \frac{e^2}{4}\right)e^{-x}$

**Q.(10)**

Let y(x) be a solution of the differential equation $(1 + e^x)y' + ye^x = 1$. If $y(0) = 2$, then which of the following statements is(are) true ?

(A) $y(-4) = 0$

(B) $y(-2) = 0$

(C) y(x) has a critical point in the interval $(-1,0)$

(D) y(x) has no critical point in the interval $(-1,0)$

**Q.(11)**

Consider the family of all circles whose centers lie on the straight line $y = x$. If this family of circles is represented by the differential equation $Py'' + Qy' + 1 = 0$, where P,Q are functions of x,y and y' (here $y' = \frac{dy}{dx}, y'' = \frac{d^2y}{dx^2}$), then which of the following statements is (are) true ?

(A) $P = y + x$

(B) $P = y - x$

(C) $P + Q = 1 - x + y + y' + (y')^2$

(D) $P - Q = x + y - y' - (y')^2$

**Q.(12)**

Let $f : [0, \infty) \to \mathbb{R}$ be a continuous function such that $f(x) = 1 - 2x + \int_0^x e^{x-t} f(t)dt$

for all $x \in [0, \infty)$. Then, which of the following statement(s) is (are) TRUE ?

(A) The curve $y = f(x)$ passes through the point (1, 2)

(B) The curve $y = f(x)$ passes through the point (2, –1)

(C) The area of the region $\{(x, y) \in [0, 1] \times \mathbb{R} : f(x) \le y \le \sqrt{1-x^2}\}$ is $\frac{\pi - 2}{4}$

(D) The area of the region $\{(x, y) \in [0, 1] \times \mathbb{R} : f(x) \le y \le \sqrt{1-x^2}\}$ is $\frac{\pi - 1}{4}$

**Q.(13)** Let $y = f(x)$ and $y = g(x)$ be functions satisfying

$$x^3 - 3x^2 f(x) + (f(x))^3 = 0, \qquad g'(x) = \frac{f(x)}{x}, \qquad g(0) = 1.$$

Then, g(x) is equal to which of the following option(s)

(A) $1 + f(x)$ (B) $x f'(x) + 1$ (C) $f'(x) + 1$ (D) $x f(x) + 1$

## NUMERICAL ANSWER TYPE QUESTIONS

**Q.(14)**

A farmer $F_1$ has a land in the shape of a triangle with vertices at P(0, 0), Q(1, 1) and R(2, 0). From this land, a neighbouring farmer $F_2$ takes away the region which lies between the side PQ and a curve of the form $y = x^n (n > 1)$. If the area of the region taken away by the farmer $F_2$ is exactly 30% of the area of $\Delta PQR$, then the value of n is ______

**Q.(15)**

Let $F(x)=\int_{x}^{x^2+\frac{\pi}{6}} 2\cos^2 t\,dt$ for all $x\in\mathbb{R}$ and $f:\left[0,\frac{1}{2}\right]\to[0,\infty)$ be a continuous function. For $a\in\left[0,\frac{1}{2}\right]$, if F'(a) + 2 is the area of the region bounded by x = 0, y = 0, y = $f$(x) and x = a, then $f$(0) is

**Q.(16)**

If y(x) is the solution of the differential equation

$$xdy-(y^2-4y)dx=0 \text{ for } x>0,\ y(1)=2,$$

and the slope of the curve $y=y(x)$ is never zero, then the value of $10y\left(\sqrt{2}\right)$ is ______.

**Q.(17)**

Let $f:\mathbb{R}\to\mathbb{R}$ be a differentiable function with $f(0)=0$. If $y=f(x)$ satisfies the differential equation

$\frac{dy}{dx}=(2+5y)(5y-2)$, then the value of $\lim_{x\to-\infty} f(x)$ is _____.

**Q.(18)** Let $A_n$ represents the area formed by the elements of set A, defined below.

$$A=\left\{(x,y)\in\mathbb{R}\times\mathbb{R}: y\le[x]+\sqrt{\{x\}}, y\ge 0, 0\le x\le n, n\in\mathbb{N}\right\}$$

If [.] denotes the greatest integer function and {.} denotes the fractional part function, then find the value of $A_3+A_6$.

**Q.(19)** Let $f(x)$ and $g(x)$ be two differentiable functions satisfying the following condition:

$$f(0)=2, g(0)=1, f(x)+g'(x)=1=f'(x)+g(x), x\ge 0.$$

The value of $2f(\ln 2)+3g(\ln 3)$ is equal to

**Q.(20)** The functions $f(x)$ and $g(x)$ are defined as

$$f(x)=\begin{cases}-2, & -3\le x\le 0\\ x-2, & 0<x\le 3\end{cases}, \text{and}$$

$$g(x)=\min\left(f(|x|)+|f(x)|,\ f(|x|)-|f(x)|\right).$$

The area bounded by $y=g(x), y=0, x=3, x=-3$, is equal to

## PASSAGES

**Passage-I:**

Let $f:[0,1]\to \mathbb{R}$ (the set of all real numbers) be a function. Suppose the function $f$ is twice differentiable, $f(0)=f(1)=0$ and satisfies $f''(x)-2f'(x)+f(x)\geq e^x$, $x\in[0,1]$.

**Q.(21)**

If the function $e^{-x}f(x)$ assumes its minimum in the interval $[0,1]$ at $x=\frac{1}{4}$, which of the following is true ?

(A) $f'(x)<f(x), \frac{1}{4}<x<\frac{3}{4}$

(B) $f'(x)>f(x), 0<x<\frac{1}{4}$

(C) $f'(x)<f(x), 0<x<\frac{1}{4}$

(D) $f'(x)<f(x), \frac{3}{4}<x<1$

**Q.(22)**

Which of the following is true for $0<x<1$ ?

(A) $0<f(x)<\infty$

(B) $-\frac{1}{2}<f(x)<\frac{1}{2}$

(C) $-\frac{1}{4}<f(x)<1$

(D) $-\infty<f(x)<0$

**Passage-II:**

Consider one side AB of a square ABCD, (read in order) on the line $y=2x-17$, and the other two vertices C, D on the parabola $y=x^2$.

**Q.(23)**

Find the minimum intercept of the line CD on y-axis.

**Q.(24)**

Find the maximum possible area of the square ABCD.

**Q.(25)**

Find the area enclosed by the line CD with minimum y-intercept and the parabola $y = x^2$.

## KEY

| | | | | |
|---|---|---|---|---|
| 1. D | 2. A | 3. A | 4. A | 5. B |
| 6. C | 7. 14 | 8. AC | 9. AC | 10. AC |
| 11. BC | 12. BC | 13. AB | 14. 4 | 15. 3 |
| 16. 8 | 17. 0.4 | 18. 24 | 19. 3.5 | 20. 11.5 |
| 21. C | 22. D | 23. 3 | 24. 1280 | 25. 10.66-10.67 |

# GR-08 : STRAIGHT LINES

## SINGLE CORRECT

**Q.(1)**

A variable line passes through P(2,3) and cuts the co-ordinates axes at A and B. If the parallelogram OACB (where O is the origin) is completed then find number of ordered pairs (x,y) of integers which lie on the locus of point C.

(A) 11 (B) 7 (C) 9 (D) 5

**Q.(2)**

Triangle ABC lies in the Cartesian plane and has an area of 70 sq. units. The coordinates of B and C are (12, 19) and (23, 20) respectively and the coordinates of A are (p, q). The line containing the median to the side BC has slope –5. Find the largest possible value of $(p+q)$.

(A) 44 (B) 45 (C) 46 (D) 47

## MULTIPLE CORRECT

**Q.(3)** Let $x+y=2$ and $2x+y=5$ are internal angle bisectors of angles B and C of ΔABC respectively. If $A(3,4)$, then which of the following options is/are correct?

(A) Image of A about internal bisector of angle B is $(-2,-1)$.

(B) If $r$ denotes inradius of ΔABC, then $2r=3\sqrt{10}$.

(C) Equation of BC is $y=3x+5$

(D) Image of A about internal bisector of angle C is $(-1,-2)$.

**Q.(4)** Let

$$L \equiv ax+by+c=0,$$
$$M \equiv bx+cy+a=0,$$
$$N \equiv cx+ay+b=0.$$

If $\lambda = a + b + c, \mu = a^2 + b^2 + c^2, \nu = ab + bc + ca; a, b, c \in \mathbb{R}$, the which of the following options is/are correct?

(A) If $\lambda = 0, \mu \neq \nu$, then the lines $L, M, N$ always pass through a fixed point.

(B) If $\lambda = 0, \mu = \nu$, then the each of L,M N are true for all points in xy-plane.

(C) If $\lambda \neq 0, \mu \neq \nu$, then then lines L,M N form a triangle.

(D) If $\lambda \neq 0, \mu = \nu$, then graph of L, M, N are same.

**Q.(5)** Let M and N be line mirrors. A ray of light along $L_1$ strikes M and reflects along line $L_2$. After second reflection from $N$, the ray moves along line $L_3$ and after third reflection from M, it moves along line $L_4$. If

$$L_1 \equiv x - 3y = 1, M \equiv y = 0 \; and \; N \equiv x = 4$$

then which of the following options is/are correct?

(A) Equation of line $L_4$ is $x + 3y - 5 = 0$

(B) Area of the figure so formed is 12 square units.

(C) Equation of line $L_3$ is $x - 3y = 7$

(D) Equation of line $L_2$ is $x + 3y = 1$.

**Q.(6)** Let $L_i; i = 1,2,3 \ldots 7$ be lines such that

$$L_1 \equiv x + y = 1, \quad L_2 \equiv x + y = 3,$$
$$L_3 \equiv 2x - y = 5, \quad L_4 \equiv 3x + y = 5.$$

Line $L_5$ passes through origin, and meets $L_1$ and $L_2$ at $P$ and $Q$ respectively. Through P and Q, lines $L_6$ and $L_7$ are drawn parallel to $L_3$ and $L_4$ respectively intersecting at point R. Which of the following option(s) can be point R.

(A) (4,3) (B) (10,4) (C) (19,7) (D) (25,10)

**Q.(7)**

Consider two isosceles triangles $\Delta AB_1C_1$ and $\Delta AB_2C_2$ such that $AB_1 = AC_1$, $AB_2 = AC_2$ and $P(1, -2)$ is point of intersection of lines $B_1C_1$ & $B_2C_2$. If A, $B_1$, $C_2$ lies on $7x - y + 3 = 0$ and A, $B_2$, $C_1$ lies on $x + y - 3 = 0$, then which of the following is/are correct ?

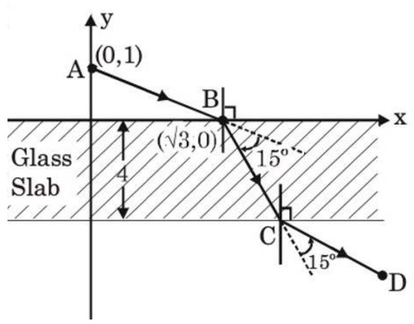
(A) point A lies on y-axis.

(B) product of x-coordinates of points $B_1, C_1, B_2, C_2$ is negative

(C) product of y-coordinates of points $B_1, C_1, B_2, C_2$ is positive

(D) sum of slope of lines $B_1C_1$ and $B_2C_2$ is $\frac{8}{3}$.

**Q.(8)** A ray of light from A to glass slab of thickness 4 units is incident at point B(as shown in figure). After refraction, it turns by 15° towards the normal and moves along BC. After second refraction it turns by 15° away from the normal, and moves along CD.




If $CD = AB$, then which of the following options is/are correct?

(A) Equation of BC is $x + y = \sqrt{3}$

(B) Coordinates of C are $(2 + \sqrt{3}, -4)$

(C) Equation of DC is $\sqrt{3}y + x = 2 - 3\sqrt{3}$

(D) Coordinates of D are $(2\sqrt{3} + 4, -5)$

**Q.(9)**

Point A lies on x-axis and point B lies on y = x. If PA.PB is minimum where P divides segment AB internally, then (given P is $(1, 2-\sqrt{3})$ and 'O' is origin), -

(A) $OA = 2\sqrt{2} - \sqrt{6} + \sqrt{3} - 1$

(B) $OB = 2\sqrt{2} - \sqrt{6} + \sqrt{3} - 1$

(C) $\angle OAB = 45°$

(D) $\angle OBA = 22.5°$

## NUMERICAL TYPE

**Q.(10)** If $\lfloor x \rfloor = \max\{a \in \mathbb{Z} : a \leq x\}$, then find the minimum value of

$$\left\lfloor \sqrt{x^2 - 2x + 2} + \sqrt{x^2 - 4x + 29} \right\rfloor.$$

**Q.(11)** Find the value of $4x$, if $\lambda$ is minimum, where

$$\lambda = AB + BC + CA; A(5,3), B(\alpha, \alpha), C(x, 0).$$

**Q.(12)**

$\Delta$ POR has vertices P(0,12), R(5,0) and O(0,0). There exist a line '$\ell$' cutting PR and OP at A and B respectively, such that circles can be inscribed in $\Delta$PAB and quadrilateral ORAB. Also these circles are tangent to the line '$\ell$' at the same point. If the area of quadrilateral ORAB is A, then find the value of $\left[\frac{A}{5}\right]$?

**Q.(13)** Find the area of pentagon formed by points lying in $\mathbb{A} \cup \mathbb{B}$, where

$$\mathbb{A} = \left\{(x, y) \in \mathbb{Z}^2 \middle| \det\left(\begin{pmatrix} x^2 & 4-y \\ y-4 & 1 \end{pmatrix}\begin{pmatrix} x-3 & y \\ y & 3-x \end{pmatrix}\begin{pmatrix} x^2+y^2+37 & 2 \\ 6x+y & 1 \end{pmatrix}\right) = 0\right\},$$

$$\mathbb{B} = \left\{(x, y) \in \mathbb{Z}^2 \middle| \det\left(\begin{pmatrix} |x-4| & -1 \\ |y-9| & 1 \end{pmatrix}\begin{pmatrix} x-6 & y-1 \\ y-1 & 6-x \end{pmatrix}\begin{pmatrix} y-5 & 7-x \\ 7-x & 5-y \end{pmatrix}\right) = 0\right\}.$$

Here $\det(A)$ denotes the determinant of matrix A.

**Q.(14)**

If value of the expression

$$\left(\sqrt{(x-2)^2+(y-4)^2} - \sqrt{(x+2)^2+(y-2)^2}\right)^4 + \left(\sqrt{(x-2)^2+(y-4)^2} + \sqrt{(x+2)^2+(y-2)^2}\right)^4$$

is minimum for $x = x_0$ and $y = y_0$, then $|x_0 + y_0|$ is equal to

**Q.(15)**

Let PQRS be a square such that point P lies on positive y-axis and Q lies on positive x-axis. If S is (7, 10) and the co-ordinate of point R is (a, b), then value of b is

## PASSAGES

Passage-I

Consider a triangle ABC.
Let $x + y - 3 = 0$, $x - 4y + 7 = 0$ and $2x - y = 0$ respectively the equations of the altitudes BE, CF and AD on the sides AC, AB and BC.

**Q.(16)**

If one of the vertex of triangle is of the form $(\alpha, 2\alpha)$, then the locus of centroid of triangle ABC as $\alpha$ varies is -

(A) $9x - 63y + 129 = 0$
(B) $9x + 63y - 129 = 0$
(C) $9x + 45y - 89 = 0$
(D) $9x + 45y - 99 = 0$

**Q.(17)**

Let Q be any point on the line $L : x - k = 0$, where k is abscissa of orthocenter of $\Delta ABC$, P is a point (k, 0) other than point Q, QR is the acute angle bisector of angle OQP, (O is origin) meets the x-axis at piont R, then the locus of foot of perpendicular from R on the OQ is -

(A) $(x^2 + y^2)(x + 2) + x = 0$
(B) $(x^2 + y^2)(x - 1) + x = 0$
(C) $(x^2 + y^2)(x - 2) + x = 0$
(D) $(x^2 + y^2)(x + 1) + x = 0$

**Passage-II**

Given four parallel lines $L_1$, $L_2$, $L_3$ and $L_4$. Let the distances between them be $d_{12}$, $d_{23}$, $d_{34}$ respectively (here $d_{ij}$ is distance between lines $L_i$ and $L_j$). Let P be a point, sum of whose distances from four lines is k $(d_{12} < d_{23} < d_{34})$.

**Q.(18)**

If $k = d_{12} + 2d_{23} + d_{34}$, then locus of 'P' is -

(A) entire region between lines $L_2$ and $L_3$.
(B) entire region between lines $L_1$ and $L_2$.
(C) entire region between lines $L_3$ and $L_4$.
(D) not possible.

**Q.(19)**

If $k = d_{12} + 2d_{23} + d_{34} + 2\alpha,\ 0 < \alpha < d_{34}$, then locus of 'P' is -

(A) entire region between lines $L_2$ and $L_3$.
(B) entire region between lines $L_1$ and $L_4$.
(C) entire region between lines $L_3$ and $L_4$.
(D) not possible.

## MATRIX MATCHING TYPE

**Q.(20)**

| Column-I | Column-II |
|---|---|
| A. In the triangle ABC, equation of the perpendicular bisector of the sides AB & AC are $x + y = 0$ & $y - x = 0$ respectively. If $A \equiv (5, 7)$ then equation of side BC is | (P) $7y - 5x = 0$ |
| B. The straight line of the family $(x + y) + \lambda(2x - y + 1) = 0$ that is farthest from $(1, -3)$ is | (Q) $y = 1$ |
| C. In triangle ABC, $A(1, 1), B(4, -2), C(5, 5)$. Equation of the internal angle bisector of angle A is | (R) $15y - 6x = 7$ |
| D. The diagonals AC and BD of a rhombus intersect at $(5, 6)$ If $A = (3, 2)$ then equation of diagonal BD is | (S) $2y + x = 17$ |

## KEY

| | | | | |
|---|---|---|---|---|
| 1. B | 2. D | 3. ABC | 4. ABCD | 5. CD |
| 6. AD | 7. ABC | 8. AD | 9. AB | 10. 6 |
| 11. 17 | 12. 5 | 13. 36.5 | 14. 3 | 15. 3 |

| | | | | |
|---|---|---|---|---|
| 16. C | 17. C | 18. A | 19. C | 20. A-P,B-R,C-Q,D-S |